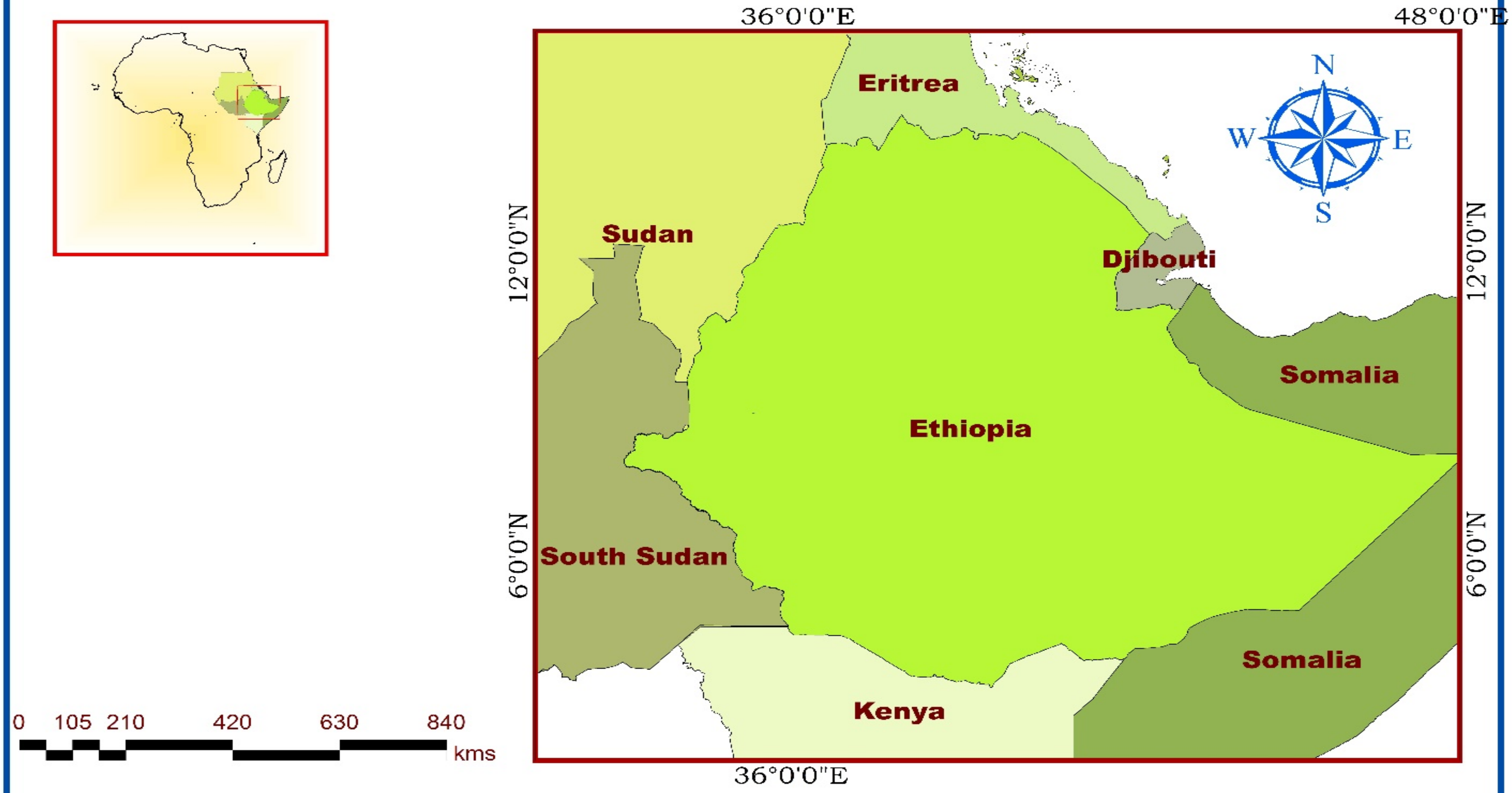
Geography of Ethiopia and The Horn



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Contents of the course

Meaning of Geography

- It is difficult to forward a definition acceptable to all geographers at all times due to the ***dynamic nature*** of the subject and ***its scope*** in its long history.
- Geography has undergone profound changes in its scope and focus:
- **In ancient times**, geography was considered to be the mother of many other sciences, including philosophy and the earth sciences.
- **In the 16th century**, geography began to emphasize location, focusing on questions of where, why and what.
- In the mid **18th century**, European geographers, especially Germans, considered the relationship between geography with philosophy.
- This approach caused geography to focus on the ***relationship between human and the natural environment.***

- In the **19th century**, geographical societies and research groups formed.
- They enhanced the role of geography as a discipline.
- In the late **20th century**, geography became ***spatial science.***
- From the ancient Greeks to modern-day geographers, geography has been **defined differently**.
- However, the following may be accepted as a ***working definition***.
- Geography is the scientific study of the Earth that describes and analyses spatio-temporal distribution, variations, interactions and relationships of phenomena (physical, biological and human) over the surface of the Earth

2. The Scope, Approaches and Themes of Geography

- Geography is **a holistic and interdisciplinary** field of study
- **Scope** means the range and variety of contents which are included in a subject or field of study.
- The scope of geography is the interface of the ***atmosphere, lithosphere, biosphere hydrosphere, and anthroposhpere.***
- *The scope is **wide and diverse in nature***

- Geography can be ***approached*** by considering *a topical-regional continuum*
- ***The topical (systematic) –approach*** *takes up particular categories of physical or human phenomena as distributed over the Earth*
- Geography of hunger, The geography of climate, The geography of agriculture, The geography of population
- ***The Regional approach-*** *focuses on concerns with the associations within regions of all or some of the elements and their interrelationships*
- The geography of Africa, Asia, or Oceania, etc

Themes of Geography

- *Geography has five basic themes*

1. **Location-** is defined as a particular place or position (Absolute and relative)

*2. **Place** refers to the physical and human attributes of a location.*

❖ *This theme is associated with toponym (the name of a place), site (the description of the features of the place), and situation (the environmental conditions of the place).*

❖ *Each place in the world has its unique characteristics expressed in terms physical and human elements*

❖ The concept of "place" aids geographers to compare and contrast two places on Earth.

3. Human-Environment Interaction

- It human-environment interaction involves three distinct aspects:
  - **Dependency**- refers to the ways in which humans are dependent on nature for a living
  - **Adaptation**- relates to how humans modify themselves, their lifestyles and their behavior to live in a new environment with new challenges
  - **Modificatio**n-allowed humans to "conquer" the world for their comfortable living and desirable end.

4. Movement - entails to the translocation of human beings, their goods, and their ideas within different spatial and temporal scale.

5. Region- A region is a geographic area having distinctive characteristics that distinguishes itself from adjacent unit(s) of space

- It could be a formal region or functional/ nodal.

1.2. Location, Shape and Size of Ethiopia and the Horn

- The **Horn of Africa**, a region of eastern Africa, is a narrow tip that protrudes into the northern Indian Ocean, separating it from the Gulf of Aden.
- It **includes** Djibouti, Eritrea, Ethiopia, and Somalia, whose cultures have been linked throughout their long history.
- In terms of size, Ethiopia is the largest of all the Horn of African countries, while Djibouti is the smallest.
- The Horn contains ***diverse physical elements*** like the highlands of the Ethiopian Plateau, the Ogaden desert, and the Eritrean and Somali coasts as well as the human aspects.

1.2.1. Location of Ethiopia

- The location of a country or a place on a map or a globe is expressed in **two different** ways.
- These are astronomical and relative locations

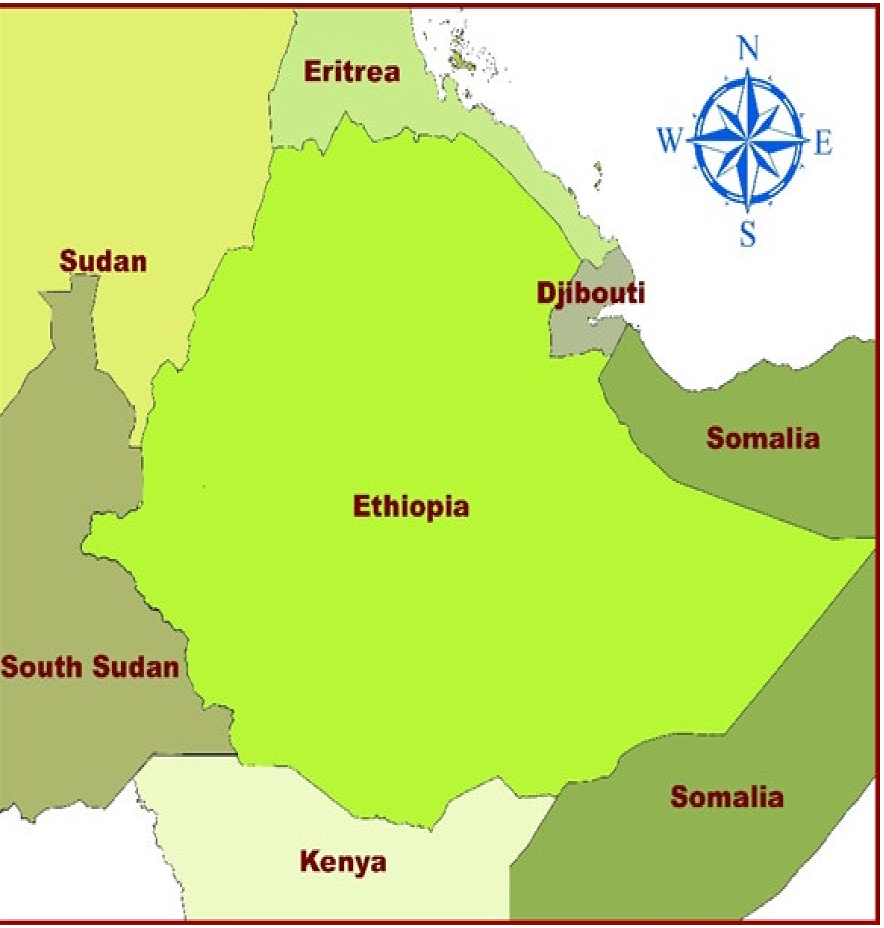
1. **Astronomical location**, also known as ***absolute or mathematical*** location, states location of places using the lines of latitudes and longitudes.

- Astronomically, Ethiopia is located between
- 3^{o}N (Moyale) and 15^{o}N (Bademe - the northernmost tip of Tigray) latitudes and
- 33^{o}E (Akobo) to 48^{o} E (the tip of Ogaden in the east) longitudes.

- The latitudinal and longitudinal extensions are important in two ways.
- **First**, as a result of its latitudinal extension the country experiences **tropical climate**
- **Secondly** due to its longitudinal extension there is a difference of **one hour** between the most easterly and most westerly points of the country

2. Relative location expresses the location of countries or places with reference to the location of ***other countries (vicinal), landmasses or water bodies.***

Relative location of Ethiopia



- Thus in relative term, Ethiopia is a ***landlocked country*** located in;
  - East and South East of Sudan
  - West of Dijbouti
  - Northeast of south sudan
  - West and nortwest of Somalia
  - South and southwest of Eritrea
  - North of Kenya

Relative location of Ethiopia

- In relation to other features in the world the location of Ethiopia might be expressed in the following way,
  - In the Horn of Africa
  - Southwest of the Arabian Peninsula
  - South of Europe
  - Northwest of the Indian Ocean
  - In the Nile Basin
- The relative location of Ethiopia has ***implications*** for its:

❖Climate Socio-cultural

❖Geopolitics Hydropolitics

1.2.2. Size of Ethiopia

- Ethiopia with a total area of approximately **1,106,000 square kilometers** is the **8th** largest country in Africa and **25th in the World**.
- The size of Ethiopia also affects both the **natural and human environment** of the country.
- The advantages and disadvantages of the size of Ethiopia are

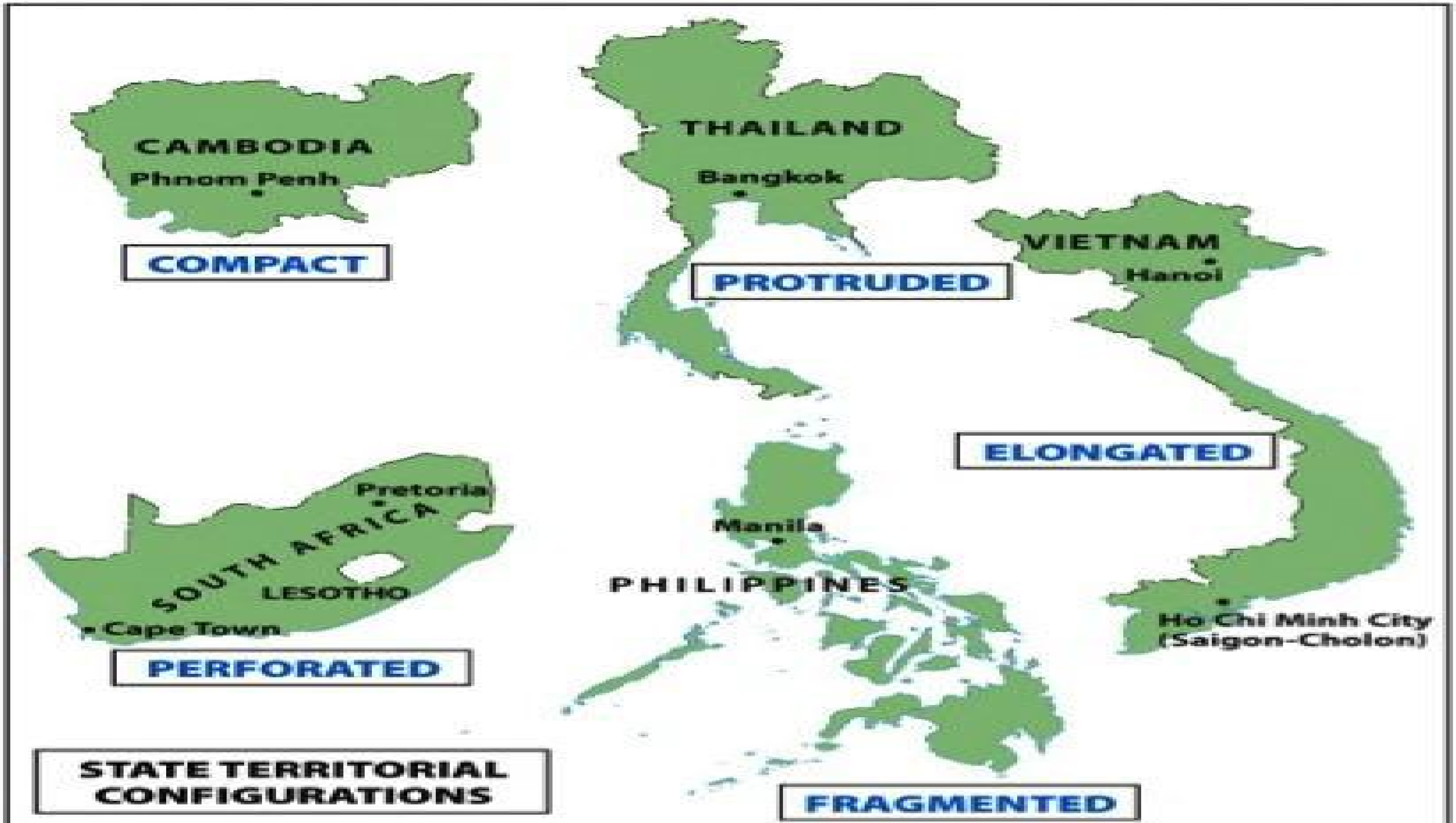
Size of Ethiopia……

Advantages	Disadvantages
• Possess diverse agro ecological zones • Variety of natural resources • Own extensive arable land • Has larger population size • Home for diverse cultures • Greater depth in defense external invasion	• Demands greater capital to construct infrastructural facilities • Requires large army to protect its territory • Difficult for effective administration • Difficult for socio-economic integration • •

1.2.3. The shape of Ethiopia and its Implication

- Countries of the World have different kinds of shape that can be divided into **five main categories**
- **Compact -** the distance from the geographic center of the state to any of the boarder does not vary greatly.
- It is easier for defense and for socio-economic integration
- **Fragmented shape countries:** They are divided from their other parts by either water, land or other countries.
- **Elongated (linear) shape countries:** are geographically long and relatively narrow like Chile.
- **Perforated shape countries:** A country that completely surrounds another country like the Republic of South Africa.

- **Protrude shape countries:** Countries that have ***one portion that is much more elongated than the rest of the country*** like Myanmar and Eritrea.Fragmented



✓There are various ways of measuring shape of countries. These measures are known as the ***indices of compactness***.

- These indices measure the ***deviation of the shape of a country from a circular shape***, which is the most compact shape.
- Since there is no country with absolutely circular shape, those approximating a circular shape are said to be more compact
- There are four most commonly used measures of compactness. These are:

1. **Area-Boundary(A/B) ratio**: The ratio of area of country to its boundary length.

- The **higher** the A/B ratio, the greater the degree of compactness.

2. **Boundary-Circumference(B/C) ratio**: The ratio of boundary length of a country to the circumference of a circle having the same area as the country itself.

- It measures how far the boundary of a country approximates the circumference of a circle of its own size.
- Therefore, the nearer the ratio to 1 the more compact the country is.

3. **Area-Circumference(A/C) ratio**. The ratio of the area of the country to the circumference of the smallest inscribing circle.

- It compares the area of the country with the circumference of a circle that passes touching the extreme points on the boundary of the country

- . The **higher** the A/C ratio, the greater the degree of compactness.

4. **Area-Area (A/A') ratio**: The ratio of the actual area of a country to the smallest possible inscribing circle

  - The area of the inscribing circle is the area of the smallest possible circle whose circumference passes through the extreme points on the boundary.
  - Half-length of the longest distance between two extreme points gives radius of the inscribing circle.
  - The **nearer the ratio to 1**, the more compact the country is.

- Ethiopia’s shape compared to some other countries

Country	Area(km^2)	Boundary (km)	A/B ratio	B/C ratio	A/C ratio
Ethiopia	1,106,000	5,260	210.27	1.41	296.61
Djibouti	22,000	820	26.83	1.56	41.83
Eritrea	117,400	2,420	48.51	1.99	96.83
Kenya	582,644	3,600	161.85	1.33	215.28
Somalia	637,657	5,100	125.03	1.80	225.22

1.3. Basic Skills of Map Reading

- A map is a simplified, diminished, plain representation of all or part of the earths surface as viewed from vertically above.
- Although many disciplines use maps, they have a special significance for Geographers as primary ***tools for displaying and analyzing spatial distributions, patterns and relations.***

Importance of maps

- Provide the basis for making **geographical details** of regions
- are powerful tools for making **spatial analysis** of geographical facts of areas represented.

Importance of maps……

- are useful for giving **location of geographical** features by varied methods of grid reference, place naming etc....
- make **storage** of the geographical data of areas represented.
- are potentially used to asses' reliable **measurements** of the geographical features. The measurements can be of ***area size, distance etc***
- are used on **various disciplines** like land use planning, military science, epidemiology, geology, economics, history, archaeology, agriculture etc.

Types of Map

- Although most maps have similar characteristics, they can differ from one another in many ways

1. **Topographical maps (General-Purpose Maps):** Topographic maps depict one or more natural and cultural features of an area.

- Show limited detailed information

- They could be small ($\leq$1:250,000), mediums (1:50,000 - 1:250,000) or large scale ($\geq$1:50,000)depending on the size of the area represented.

2. Special purpose/statistical maps/thematic maps or topical maps

- These are maps, which show distribution of different aspects such as temperature, rainfall, settlement, vegetation etc.
- Soil maps Vegetation maps Climate map

Marginal Information on Maps

- Maps are used to convey information.
- To read maps effectively, map users need information about the map.
- Such information is presented in the maps margins and is known as marginal information.
- Marginal information includes:

a. **Title:** It is the heading of the given map which tells what the map is all about.

b. **Key (legend):** It is the list of all convectional symbols and signs shown on the map with their interpretation.

c. **Scale:** It is the *ratio* between the distance on the map and the actual ground distance.

•This information indicates the extent to which the area that is represented in the map has been ***reduced***

• Scales enable the map user to interpret the ground measurement like road distance, areal sizes, gradient etc.

- It can be expressed as *representative fraction, statements/verbal scale, and linear (graphic) scale*.

d. **North arrow:** It is indicated with the north direction on a map; used to

- Shows the north direction on the map.

e. **Margin:** Is the frame of the map. It is important for showing the end of the mapped area.

f. **Date of compilation:** It is a date of map publication.

- This enables map users to realize whether the map is updated or out-dated.

- **Basic Principles of Map Reading- map readers need to have**
  - ideas about the symbol and also the real World (landscapes)
  - knowledge of directions is an important principle in reading maps
  - locating *places using latitudes and longitudes or grid references*

✓ left to right,- easting (heading eastward), and

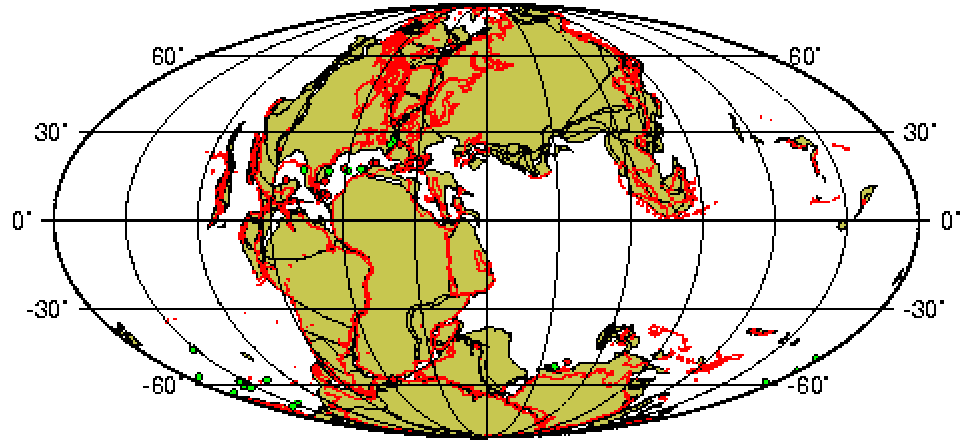
✓northing's (heading in a northward direction).

Chapter Two: The Geology Of Ethiopia And The Horn

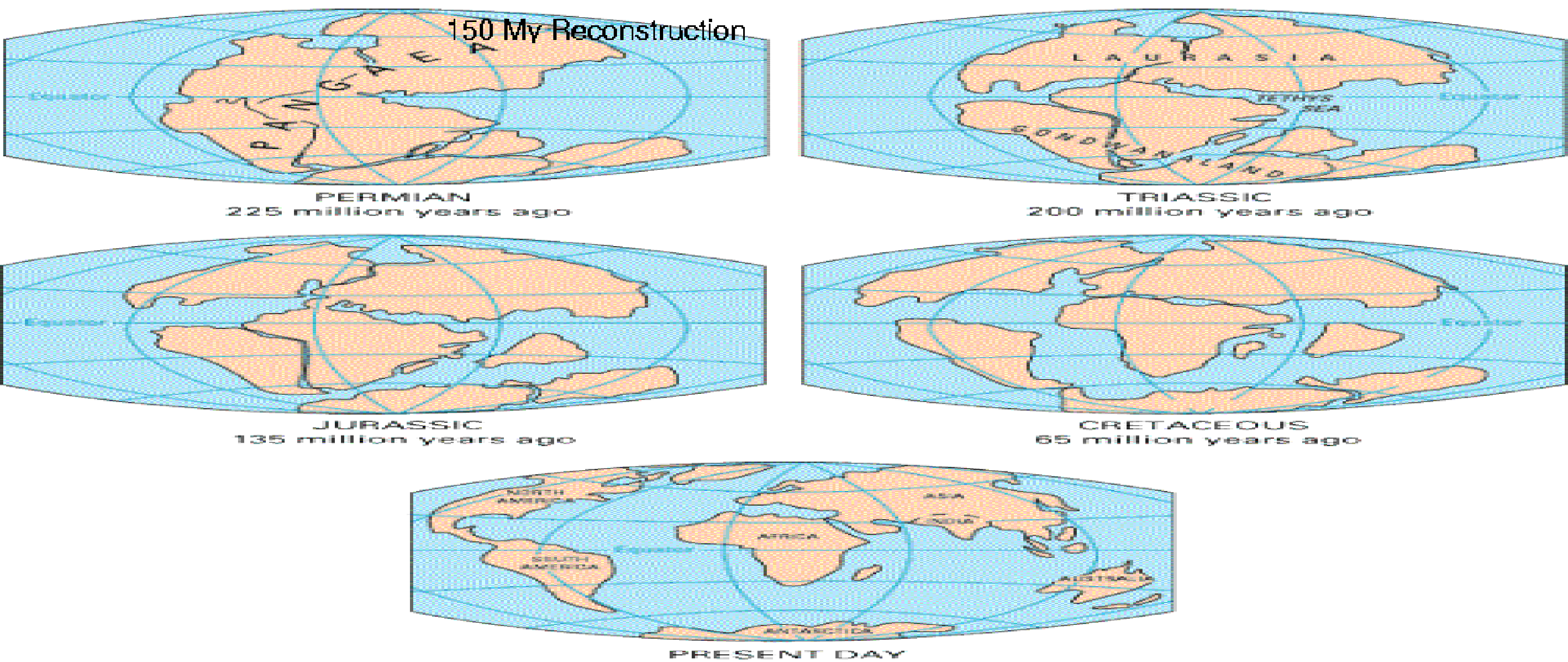
- ✓ **Geology** is an **Earth science** that studies the evolution of the earth, the materials of which it is made of and the processes acting upon them.
- ✓ Much of Geology is concerned with **events that took place in the remote past**
- ✓ Geological understanding must, therefore, be obtained by **inferences and clues**
- ✓ These clues are based on
  - Rocks and landforms- Direct Method
  - Characteristics of seismic waves – **Geo-Physics**
  - **Geochemistry** (analysis of the detailed composition of rocks which can give clues as to their origin)
  - **Geochronology** (methods for finding the ages of rocks, usually from the radioactive elements they contain).

Continental Drift Theory

- Alfred Wegener proposed the hypothesis of ***continental drift in 1911***
- The continental drift hypothesis proposes that the earth's continents were once assembled to form the super single continent **Pangaea**.
- The large super continent was then split into **Gondwanaland** where Africa is a part and **Laurasia**; and later into smaller fragments over the last million years.
- These then drifted apart to form the present arrangement of continents.



150 My Reconstruction

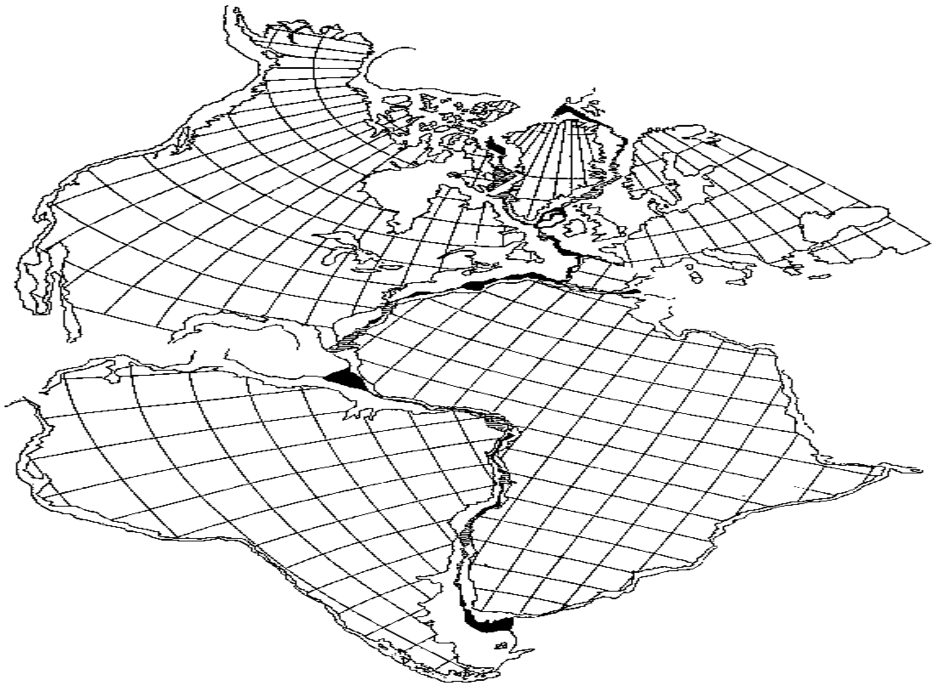


Wegener's Evidence

He gathered information from many different sources and used it as evidence for his hypothesis

1. Structural- *Fit of the continents:*

- The opposing coastlines of continents often fit together like puzzle pieces

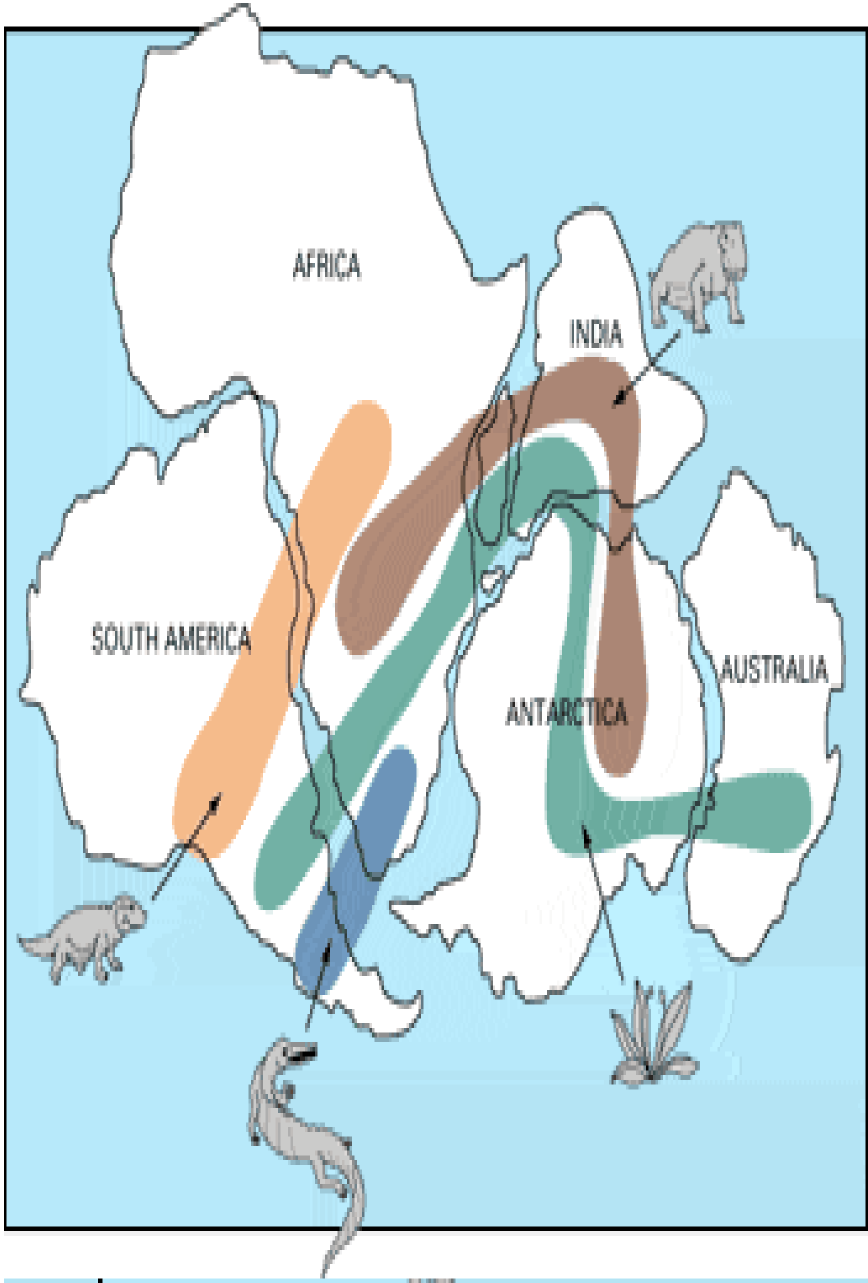


2. Palaeontological- Distribution of fossils:

•The distribution of plants and animal fossils on separate continents forms definite linked patterns if the continents are reassembled.

•*fossils are remains of living things that lived long ago.*

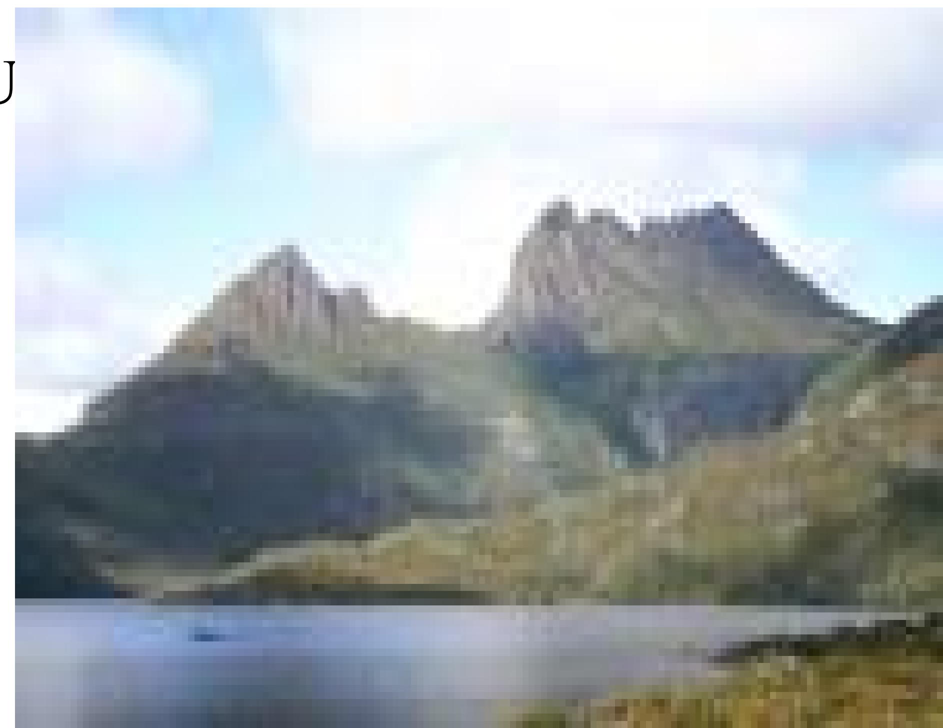
•similar fossils have been discovered in matching coastlines on different continents.



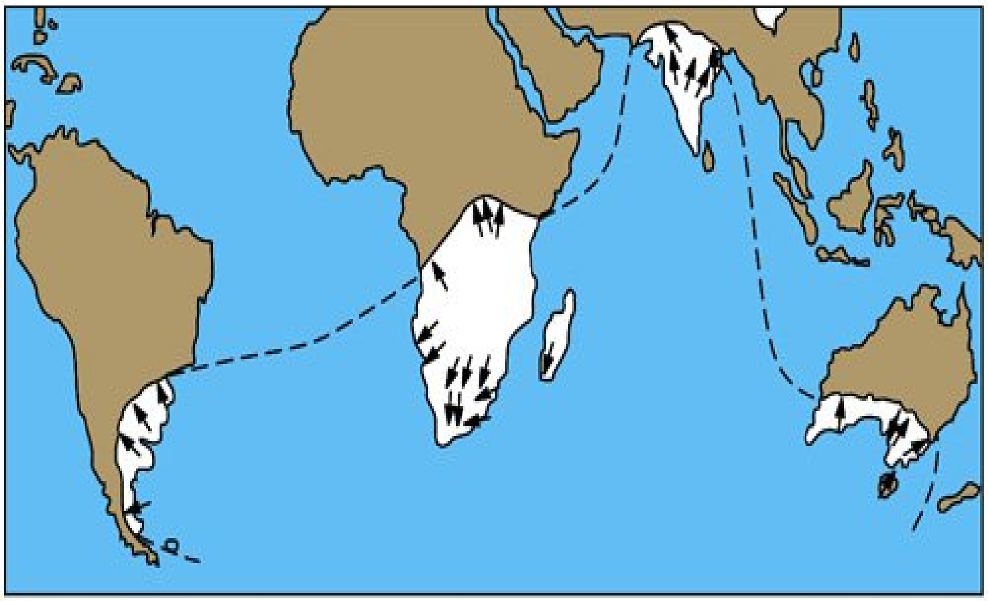
Wegener's Evidence……

3. Match of mountain belts, rock types: If the continents are reassembled as Pangaea, mountains in West Africa, North America, Greenland, and Western Europe match up.

Some mountain ranges on different continents seem to match Ex: ranges in Canada match Norway and Sweden



4. Paleoclimatic- : rocks formed 200 million years ago in India, Australia, South America, and southern Africa all exhibited evidence of continental glaciations.



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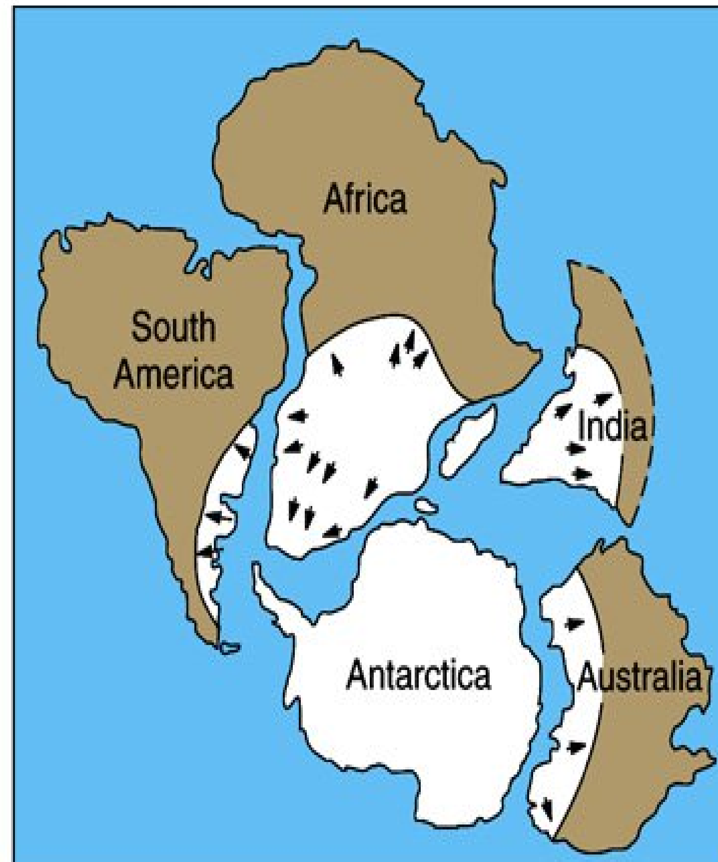

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2.1 The Geologic Processes: Endogenic & Exogenic Forces

- ✓ Geology studies how Earth's materials, structures, processes and organisms have ***changed over time***.
- ✓ **These processes are divided into two major groups**

1. The internal processes (endogenic) include volcanic activity and all the tectonic processes (folding, faulting, orogenesis (mountain building), and epeirogenesis (slow rising and sinking of the landmass).

- These processes result in building of structural and volcanic features like plateaus, rift valleys, Block Mountains, volcanic mountains, etc.

2. The external (exogenic) processes are geomorphic processes.

- They include **weathering, mass transfer, erosion and deposition**.
- They act upon the volcanic and structural landforms by modifying, roughening and lowering them down.
- The landmass of Ethiopia, as elsewhere, is the result of the combined effect of endogenic and exogenic processes

2.3. The Geological Time Scale and Age Dating Techniques

- The Earth is believed to have been formed approximately **4.6 billion** years ago and the earliest forms of life were thought to have originated approximately **3.5 billion** years ago.
- The geological time scale measures time on a scale involving **four main units:**
- **An** ***epoch*** is the smallest unit of time on the scale and encompasses a period of millions of years.
- Epochs are clumped together into larger units **called** ***periods.***
- Periods are combined to make subdivisions called ***Eras*****.**
- An ***eon*** is the largest period of geological time.

Era	Period	Began (in Million Years)	End	Major Events (million years ago)
Cenozoic	Quaternary	1.6	Present	Major glaciers in North America and Europe (1.5)
	Tertiary	70	1.6	Rocky Mountains (65), individual continents take shape.
Mesozoic	Cretaceous	146	70	Dinosaurs extinct (65), western interior seaway and marine reptiles (144 – 65)
	Jurassic	208	146	Pangaea (one land mass) begins to break up (200)
	Triassic	225	208	First mammals and dinosaurs
Paleozoic	Permian	290	225	Greatest extinction on Earth (245)
	Pennsylvanian	322	290	First reptiles
	Mississippian	362	322	Coal-forming forests
	Devonian	408	362	First land animals and first forests (408)
	Silurian	439	408	Life invades land
	Ordovician	510	439	First fish appeared
	Cambria	600	510	Great diversity of marine invertebrates
Precambrian	Proterozoic	2,500	600	Marine fossil invertebrates (600)
	Archean	4,500	2,500	Earliest fossils recorded (3,500), earlies t rock formation (4,000)

Age Dating Techniques

- There are two techniques of knowing the age of rocks: **Relative and absolute age dating.**
- **Relative dating is** geological evidence to assign ***comparative ages of fossils***.
- Use the layers of rock formation and sequences of depositions –

✓"What is on top of the older rocks?“

- This only works if the area has been ***undisturbed***.

- **Absolute Dating-** Also known as **Radiometric** techniques.
- Uses radioactive elements such as uranium (U) and thorium (Th) decay naturally to form different elements or isotopes of the same element.
- Every radioactive element has its own **half-life**.
- **Half-life** is the time needed for half of a sample of a radioactive element to undergo radioactive decay and form daughter isotopes.
- Two of the major techniques include:

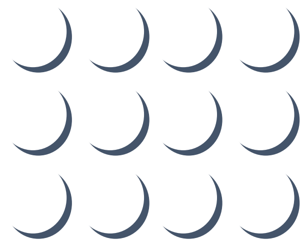
1. Carbon-14 Technique*:* Upon the organism's death, carbon-14 begins to disintegrate at a known rate, and no further replacement of carbon from atmospheric carbon dioxide can take place.

- Carbon-14 has half-life of 5730 years.

***2. Potassium-Argon Technique:* (K-Ar) Dating**

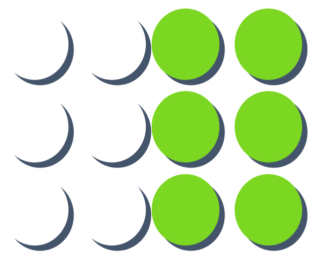
- The decay is widely used for dating rocks.
- Geologists are able to date entire rock samples in this way, because potassium-40 is abundant in micas, feldspars, and hornblendes.
- Potassium-40 has half-life of 1.3 billion years.

Rate of Decay



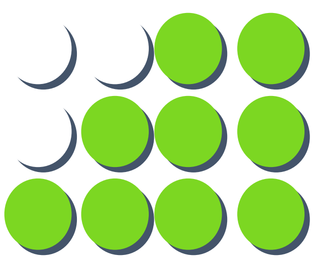
t 0

All atoms are parent isotope or some known ratio of parent to daughter



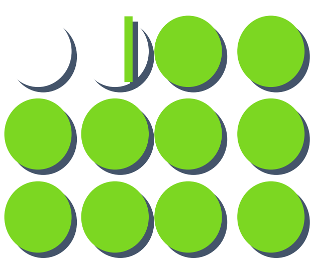
t 1

1 half-life period has elapsed, half of the material has changed to a daughter isotope (6 parent: 6 daughter)



t 2

2 half-lives elapsed, half of the parent remaining is transformed into a daughter isotope (3 parent: 9 daughter)



t 3

3 half-lives elapsed, half of the parent remaining is transformed into a daughter isotope (1.5 parent: 10.5 daughter)

We would see the rock at this point.

2.4. Geological Processes and the Resulting Landforms of Ethiopia and the Horn

2.4.1. The Precambrian Era Geologic Processes (4.5 billion - 600 million years ago)

- It covers **5/6th** of the Earth's history.
- Due to its **remoteness in time** and the **absence of well-preserved fossils**, our **knowledge of the events is limited**.
- The major geologic event of this Era was **Orogenesis**-that resulted in intense folding
- Old basement complex (crystalline basement= metamorphic rock= Precambrian era)
- In addition, the Precambrian era was known for extensive **denudation** responsible for **peneplanation**.
- The penepnained land was covered with younger rock formation later in the Mesozoic and cenozoic eras.

The Precambrian Era Geologic Processes ....

- These sequences of geological processes and activities have resulting landforms and structures.
- As such, the Precambrian rocks are overlaid by later rock formations of different ages.
- Despite the Precambrian rocks are overlaid by sequences of rock formation, they are **fond exposed** in the following areas:
  - **In the northern part**: Western lowlands, parts of northern and central Tigray.
  - **In the western Part:** Gambella, Benishangul-Gumuz (Metekel and Asossa), western Gojjam, western Wellega, Illuababora, and Abay gorge.
  - **In the southern Part:** Guji, southern Omo, and parts of southern Bale and Borena.
  - **In the eastern part:** Eastern Hararghe.

2.4.2. The Paleozoic Era Geologic Processes (600million - 225 million years ago)

- The major geological process of this Era was **denudation (erosion).**
- The gigantic mountains that were formed by the Precambrian orogeny were subjected to **intense and prolonged denudation**.
- At the end, the once gigantic mountain ranges were **reduced to a "peneplained" surface.**
- **Because of the limited deposition within Ethiopia, rocks belonging to this Era are rare in the country**

2.4.3. The Mesozoic Era Geologic Processes (225-70 million yrs ago)

- It was an Era of slow sinking and rising (**epeirogenesis**) of the landmass.
- This process affected the **whole present-day Horn of Africa and Arabian landmass.**
- At the same time the land was **tilted eastward** and therefore **lower in the southeast and higher in the northwest.**
- The **three periods** of the Mesozoic era experienced epeirogenesis in one form or another.

Triassic period: During this period, land subsidence (sinking) began in the south eastern part of Ethiopia and progressed towards the north western part of the country.

- Sinking was followed by the ***sea invasion (transgression)*** from the Indian Ocean.
- As a result, the sea invaded the land from the ***southeast towards the northwest***
- Transgression of the sea resulted in the ***deposition (sedimentation***) of the first sedimentary rock in Ethiopia which is now called **Adigrat (lower) sandstone**, after its geological naming.

- Due to the **tilting of the landmass** during the transgression and regression of the sea, and **due to the direction of the invading and retreating sea,** the age and thickness of the Sandstone layers vary in a Southeast - Northwest direction
- As such, the Adigrat sandstone is ***older and thicker in the southeast*** and progressively decreases in age and thickness north-westward

Jurassic period

- It is manifested by the **continued sinking** of the land and deepening of the sea in Ethiopia.
- As a result, **sedimentation of another** (second type of sedimentary rock) layer took place.
- This is now called **Hintalo limestone**.
- This limestone is composed of remains of ***dead creatures*** of the invading sea, sands and gravels.

Cretaceous period

- During this period great up lifting began.
- In Ethiopia, great uplifting took place in the north western part. Because of this, the sea which had covered the land surface of Ethiopia was forced to go back (regress).
- The regression was followed by the deposition of the third layer, called **upper sandstone**, which is thicker and younger in the Southeast, while in the Northwest it is older and thinner.

- The transgressing sea and Mesozoic sediments nearly covered the **whole of Ethiopia**.
- Hence, by the end of the Mesozoic era, many parts of Ethiopia were ***covered by the three layers of Mesozoic marine sediment.*** Such as

❖ ***Central Tigray***,

❖ ***western slopes of Western highlands,*** and

❖ ***Southeast lowlands.***

2.4.4. The Cenozoic Era Geologic Processes (70million years ago - Present)

- The **tectonic and volcanic activities** that took place in this Era have an important effect in the making of the present-day landmass of Ethiopia and the Horn of Africa.
- These five major geologic activities are:

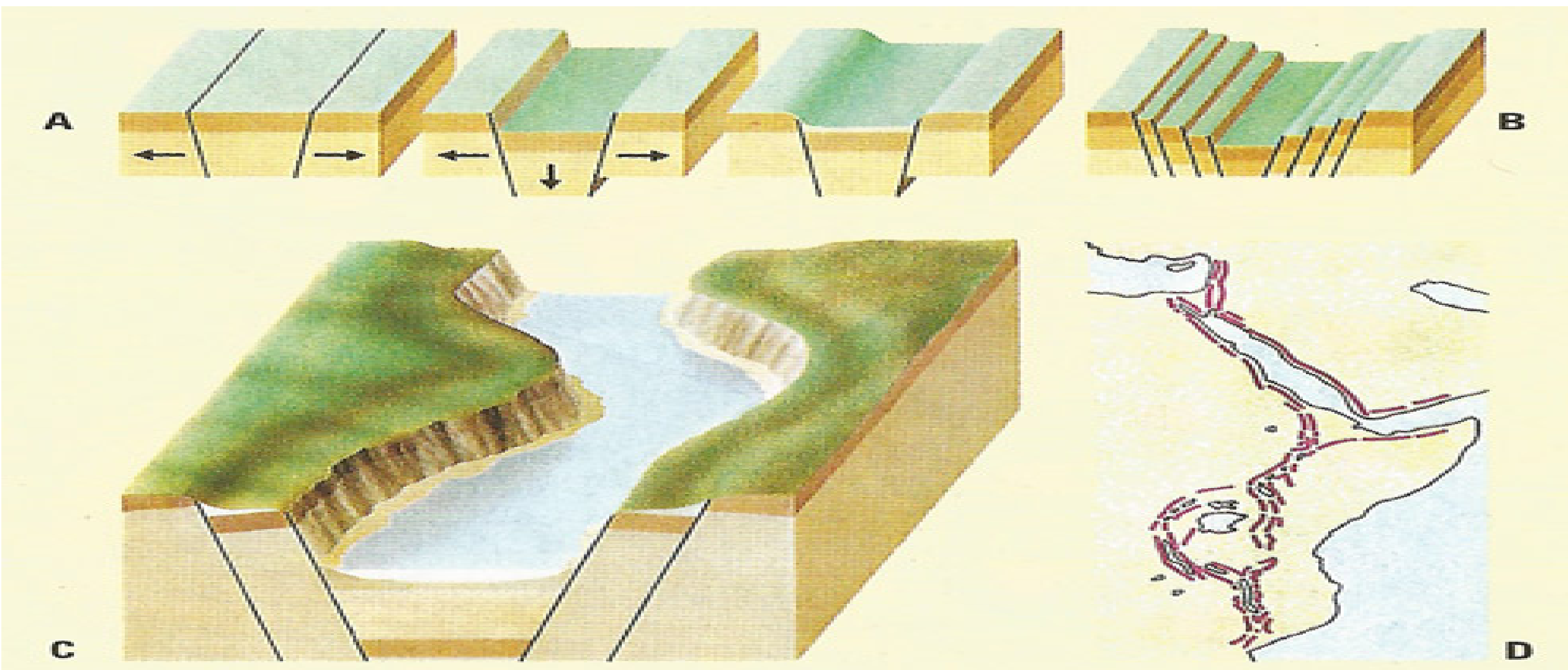
A) The continuation of the ***uplifting of land masses*** that began at the end of the Mesozoic era resulted in ***elevated surface*** mainly in central Ethiopia reaching a maximum height of 2000meters.

B) Following this uplifting and fracture, ***a huge mass of lava*** known as the Trappean lava series flowed out covering the Mesozoic sediment deposits ***forming the plateaus of Ethiopia***.

C) Major faulting and rifting took place in Central Ethiopia and **Ethiopian rift system** was formed.

- It is said to be related with the **theory of plate tectonics.**
- According to the theory, the Rift Valley may be lying on the Earth's crust below which ***lateral movement of the crust in opposite directions producing tensional forces*** (in the divergent plate boundaries) that caused parallel fractures or faults on the sides of the up-arched swell.
- As the tension widened the fractures, ***the central part of the landmass collapsed*** to form an extensive structural depression known as **the Rift Valley**

- ***Reversed tilting and volcanic activity***, later (Pleistocene) blocked the connection and isolated the extension of the sea, allowing much of the ***water to evaporate***.
- As a result, ***thick saline materials*** accumulated.



- During the same period, the area between the ***Danakil Depression*** and the ***Red Sea*** was uplifted to form the ***Afar Block Mountains***.
- The **Ethiopian Rift Valley** is part of the Great East African Rift system that extends from ***Palestine-Jordan in the north to Malawi-Mozambique*** in the south, for a distance of about 7,200 kilometers. Of these, 5,600 kilometers is in Africa, and 1,700 kilometers in Eritrea and Ethiopia.
- On land, the widest part of the Rift Valley is the ***Afar Triangle*** (200-300 km).

- The **Red Sea, the Gulf of Aden, and the East African System meet** and form the triangular depression of the Afar where the ***Kobar Sink*** lies about 125 meters below sea level.
- The Rift Valley region of Ethiopian is the most ***unstable*** part of the country.
- There are numerous ***hot springs, fumorales, active volcanoes, geysers, and frequent earthquakes.***
- The formation of the Rift Valley has the following structural (physiographic) effects:

✓It divides the Ethiopian Plateau into two.

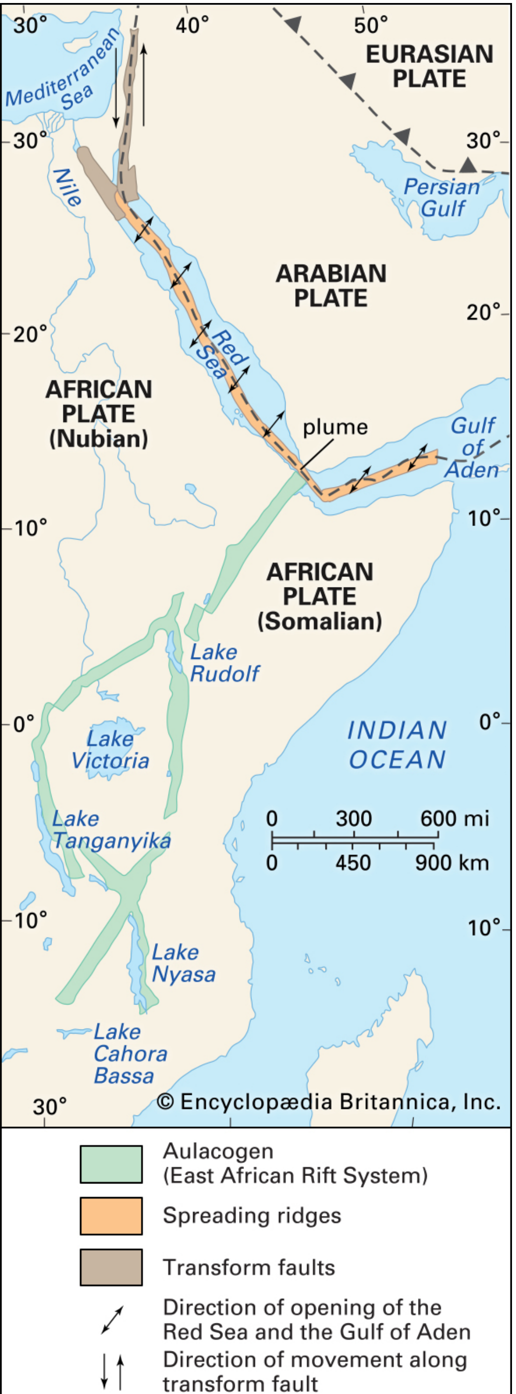
✓It separates the Arabian landmass from African landmass.

✓It causes the formation of the Dead Sea, Red Sea and the Gulf of Aden troughs.

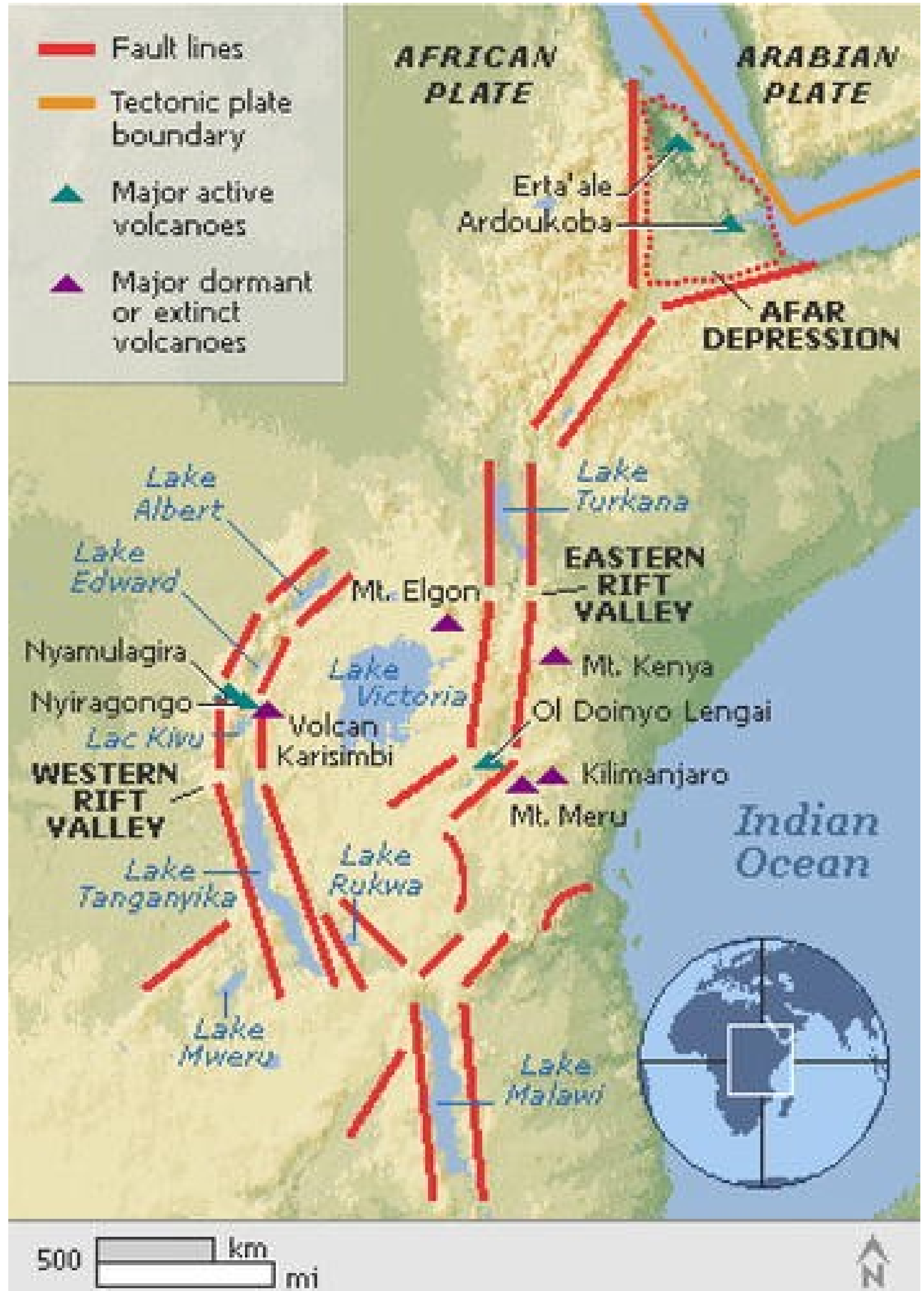
✓It creates basins and fault depressions on which the Rift Valley lakes are formed.

- Faulting and graben formation are ***not only limited to the Rift Valley***. For example, similar tectonics activities have occurred in the ***Lake Tana Basin.***
- However, the formation of Lake Tana had been accentuated by ***volcanic activity*** so that lava flow in the southeast had dammed part of the rim to deepen the basin


16°N
12°N
8°N
4°N
32°E
36°E
40°E
44°E
48°E
km
0
500
m
3000
2000
1000
0



EURASIAN PLATE
Mediterranean Sea
Nile
Persian Gulf
ARABIAN PLATE
Red Sea
AFRICAN PLATE (Nubian)
plume
Gulf of Aden
AFRICAN PLATE (Somalian)
Lake Rudolf
Lake Victoria
INDIAN OCEAN
Lake Tanganyika
0 300 600 mi
0 450 900 km
Lake Nyasa
Lake Cahora Bassa
© Encyclopædia Britannica, Inc.
Aulacogen (East African Rift System)
Spreading ridges
Transform faults
Direction of opening of the Red Sea and the Gulf of Aden
Direction of movement along transform fault



Fault lines
Tectonic plate boundary
Major active volcanoes
Major dormant or extinct volcanoes
AFRICAN PLATE
ARABIAN PLATE
Erta'ale
Ardoukoba
AFAR DEPRESSION
Lake Albert
Lake Turkana
Lake Edward
Mt. Elgon
EASTERN RIFT VALLEY
Nyamulagira
Mt. Kenya
Lake Victoria
Nyiragongo
Volcan Karisimbi
Ol Doinyo Lengai
Lac Kivu
WESTERN RIFT VALLEY
Kilimanjaro
Mt. Meru
Indian Ocean
Lake Tanganyika
Lake Rukwa
Lake Mweru
Lake Malawi
500 km
mi
N

D) The deposition of quaternary volcanic eruption and depositions are often termed as the **Aden volcanic series** which includes:

- Numerous and freshly preserved **volcanic cones**, many of which have **explosive craters**.
- Some of these are active **Dubi, Erta Ale, Afrera** etc. Of these, **Erta Ale is the most active volcano** in Ethiopia.
- **Volcanic hills and mountains**, some of which are semi-dormant (Fantale, Boseti-Gouda near Adama, Aletu north of Lake Ziway, Chebbi north of Lake Hawassa etc.).
- **Extensive lava fields and lava sheets** some of which are very recent.
- **Lava ridges**.
- **Thermal springs, fumaroles** etc.

E. The occurrence of marked climatic variations in the quaternary period resulted in quaternary sediments/recent deposits.

- During the Quaternary period of the Cenozoic Era, the Earth experienced a **marked climatic change**, **where warmer and dry periods were alternating with cooler and wet periods**..
- This was the time of the last "**Ice Age**" in the middle and high latitude areas and the time of the "Pluvial Rains" in Africa.
- The heavy Pluvial Rains **eroded the Ethiopian plateau** and the eroded materials were **deposited in the Rift Valley lakes**.

- Lake and marshy areas became **numerous and deep**: Many were enlarged and covered much area and even merged together.
- Example, Ziway-Langano-Shalla; Hawasa-Shallo; Chamo-Abaya; Lake Abe and the nearby smaller lakes and marsh basins formed huge lakes
- After the "Pluvial Rains", the Earth's climate became **warmer and drier.**
- Thus, it **increased the rate of evaporation that diminished the sizes of the lakes.**
- Areas covered with water and sediments were exposed.
- Accordingly, there are different types of deposits which are known to us at present.

- These deposits are divided into the following based on ***the place they occupy, the manner of deposition and depositing agent***.

❖ *Lacustrine deposits:* Deposits on former lakebeds and swampy depressions.

❖ *Fluvial deposits:* Deposits on the banks of rivers, flood plains both in plateau, foothills etc.

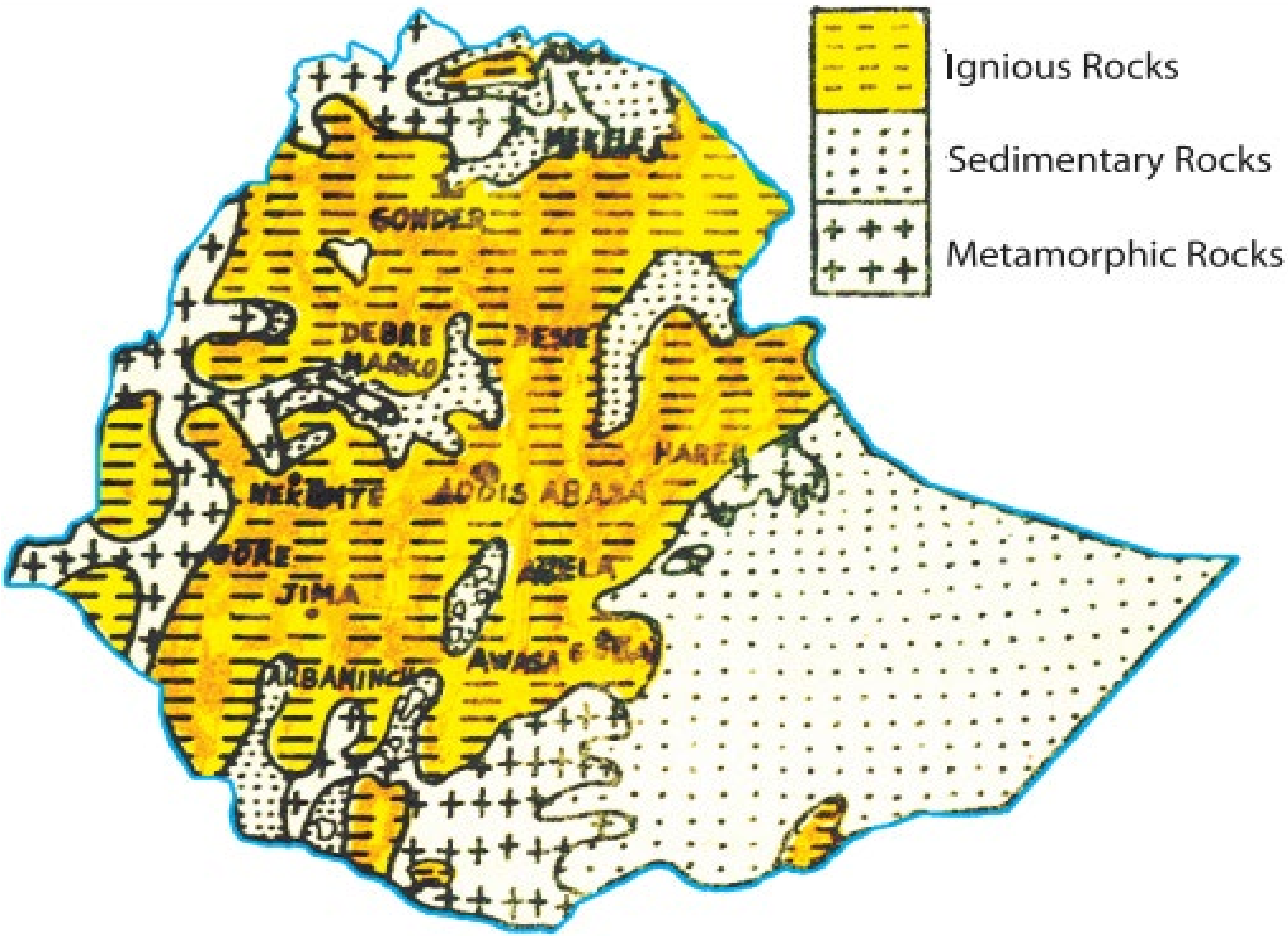
❖ *Glacio-fluvial deposits and erosional features:* These occurred on high mountains, such as Bale and Kaka Mountains.

❖ *Aeolian deposits:* are windblown deposits.

❖ *Coastal and marine deposits:* Deposits on sea invaded and sea-covered places.

- The quaternary deposits are mainly found in the Rift Valley (Afar and Lakes Region), Baro lowlands, southern Borena, and parts of north western low lands.
- Generally, the Cenozoic rocks cover ***50% of the land mass of the country.***
- These include ***Highland Tertiary volcanic (basalts), Tertiary as well as Quaternary volcanic, and sediments of the rift valley***.

2.5. Rock and Mineral Resources of Ethiopia



- **The occurrence of metallic** minerals in Ethiopia is associated with the **Precambrian rocks**.
- The exploitation and search for mineral deposits in Ethiopia has been taking place **for the past 2,000 years**
- Such has been the case of **gold production** and utilization, which has become part of Ethiopia's history, tradition and folklore (legend).
- The mining and working of **iron** for the manufacture of tools, utensils and weapons, and the use of **salt and salt-bar** all these indicate to a fairly **long mining tradition**.
- However, presently mineral production from Ethiopia has been **negligible by World standards**.

2.5.1. Brief Facts and Current State of Main Minerals in Ethiopia

- **Geological** surveys proved that Ethiopia has abundant mineral resources of metals and precious metals, coal, and industrial minerals.
- **Gold** has been mined in Ethiopia for quite long time, mainly from Benishangul-Gumuz (Metekel) and Adola.

▪ Operating mines produce gold from **primary sources** in such localities as Dermi-dama,

- Sakoro and
- Lega-dembi.

- Mechanised alluvial working is confined to the state operated gold field of Adola.

- **Secondary gold deposits** are common in the following localities:

✓Adola, Murmur Basin,

✓Makanisa (Guba and Wombera), Kaffa. Shakiso, Awata Basin, Dawa Basin, Ghenale Basin, Ujama Basin,

✓ In Gambella and Illuababora (Akobo River),

✓In Sidama (Wondo),

✓Borena (Negele-Yabelo area) and

✓in Benishangul-Gumuz (Sherkole),

- West Wellega, Mengi-Tumat-Shangul areas to the Sudanese border, and the drainage of the Didessa and Birbir.

Platinum: The ***Yubdo area in Wellega***, is the only active Ethiopian Platinum mine. Platinum occurrences have been reported from Delatti in Wellega, and the valley of Demi-Denissa and Bone Rivers as well as Tullu Mountain area in Sidama.

Tantalum: Significant deposit of tantalum and niobium is found in ***southern Ethiopia.***

✓It occurs in Adola area where Kenticha Tantalum mine with resources of more than 17,000 metric tons of world class ore reserve is found.

- The **sedimentary and volcanic rock** activities are also resourceful.
- Extensive lignite deposits in Ethiopia are found in Nedjo (Wellega), and in small amounts in Chilga (Gonder) are found in the sedimentary formations laid in between Trapean lava.

- However, **important Lignite,** one of the **lowest ranked coal**, is known to occur in many localities such as in the Beressa Valley and Ankober (North Shewa), Sululta (nearAddis Ababa), Muger Valley (West Shewa), Aletu valley (near Nedjo), Kariso and Selmi Valleys (Debrelibanos), Zega wodem gorge (near Fiche), Didessa Valley (southwest of Nekemte), Kindo and Challe Valley (Omo confluence), Adola,Wuchalle (north of Dessie), Chukga area (on Gonder-Metema road), Dessie area (near Borkena River).
- These areas are promise to be a good prospect to meet some of the local industrial and domestic needs.

Gemstones

- Gemstones, including amethyst, aquamarine, emerald, garnet, opal, peridot, sapphire, and tourmaline occur in many parts of Ethiopia, mainly in **Amhara and Oromia Regional States.**
- Quality Opal was first discovered by local people in Wadla and Dalanta woredas, North Wello in Amhara Regional State.

Potash

- The potash reserve in the Danakil (Dallol Depression) of the Afar region is believed to be significant.

Gypsum and Anhydrite

- ***A limited amount of gypsum is produced*** for domestic consumption in Ethiopia, mainly for the cement industry, ***but very large deposits are known to occur*** in sedimentary formations of the Red Sea coastal area, Danakil Depression, Ogaden, Shewa, Gojjam, Tigray, and Hararghe.
- Total reserves are probably enormous because the thickness of the gypsum deposits is many hundreds of meters and the formation are known to extend laterally for hundreds of kilometers.

Clay

- Ethiopia is endowed with industrial clay material.
- Alluvial clay deposits for bricks and tile, pottery and pipe industry occur in Adola, Abay gorge, and the Rift Valley lakes region.
- Ceramic clay for the production of glasses, plates, bricks is found at Ambo and Adola.
- Tabor ceramic industry in Hawassa gets most of its raw materials from local sources.

Marble

- Crystalline limestone is widespread in the basement rocks of Ethiopia.
- Marble has been quarried in such localities as west of Mekelle and south of Adwa in Tigray.
- In the east in Galetti, Soka, Ramis, Rochelle, Kumi and other valleys of Chercher Mountain in West Hararghe.
- In the northwestern also in areas built of Precambrian schist in Gonder, and the Dabus River and other neighboring river basins in Benishangul-Gumuz and Gojjam.

- **Construction stones**
- **Basalt, granite, limestone and sandstone** are important building stones.
- For the surfacing of roads and compaction, basalt, scoria and other volcanic rocks are extensively used.
- Mesozoic limestone is an important raw material for cement and chalk production.
- The earlier cement works at Dire Dawa and the recent ones at Muger Valley, Abay gorge (Dejen), Tigray (Messebo) are using similar raw materials from these rock formations.

2.5.2. Mineral Potential Sites of Ethiopia

- According to the Ethiopian geological survey, the geologic formations that host **most mineral potentials of Ethiopia** includes **three major greenstone** belts and other formations. These are:

1. The Western and South-western-greenstone belt

- They contain various minerals: ***primary gold occurrences*** (Dul,Tulu-Kape,Oda-Godere, Akobo,Baruda,Bekuji-Motish and Kalaj);Yubdo Platinum, Base metals of AzaliAkendeyu, Abetselo and Kata;Fakushu Molybdenite and the ***iron deposits*** of Bikilal, Chago, Gordana and Korre, Benshagul-Gumuz- Marble, Akobo and Asosa placer gold deposits and etc.

2. The Southern greenstone belt:

- It is known as the ***Adola belt***, which comprises the primary ***gold deposits*** and occurrences of Lega-dembi, Sakaro, Wellena, Kumudu, Megado-Serdo, Dawa Digati, Moyale and Ababa River; the columbo-tantalite of kenticha and Meleka, and the Adola nickel deposit and other industrial minerals.

3. The Northern greenstone belt(Tigray):

- This belt comprises of the ***primary gold occurrences*** of Terakemti, Adi-Zeresenay, and Nirague.
- The base metals of Terer, Tsehafiemba and other parts of Tigray, Placer gold occurrences of Tigray.

Chapter 3: The Topography Of Ethiopia And The Horn

3.1 General Characteristics of the Ethiopian Physiography

- ***Flat-topped plateaus, high and rugged mountains, deep river gorges and vast plains***.
- Altitude ranges from 125 meters below sea level (**Kobar Sink**) to the highest mountain in Ethiopia, **Mount Ras Dashen** (4,620 m.a.s.l), which is the fourth highest mountain in Africa.
- **The Roof of East Africa** because more than 50% of the Ethiopian landmass is above 1,000 meters of elevation; and above 1,500 meters makes 44% of the country.
- For most parts of the country are sources of many rivers and streams, the country is described as the "**Water Tower of East Africa**".

- Taking the ***1,000 meters*** contour line for the highland-lowland demarcation, one observes the following contrasting features between the Ethiopian highlands and lowlands:

- **Characteristics of Ethiopian highlands**:

  - Moderate and high amount of rainfall (>600 mm per year).
  - Lower mean annual temperature (<**20**^{0}C).
  - The climate is favorable for biotic life.
  - Rain-fed agriculture is possible.
  - Free from tropical diseases.
  - Attractive for human habitation and densely settled.

- Highlands make up nearly 56% of the area of the Ethiopia.
- Highlands further subdivided into lower highland (1,000 - 2,000 m.a.s.l) and higher highland (>2,000 m.a.s.l).

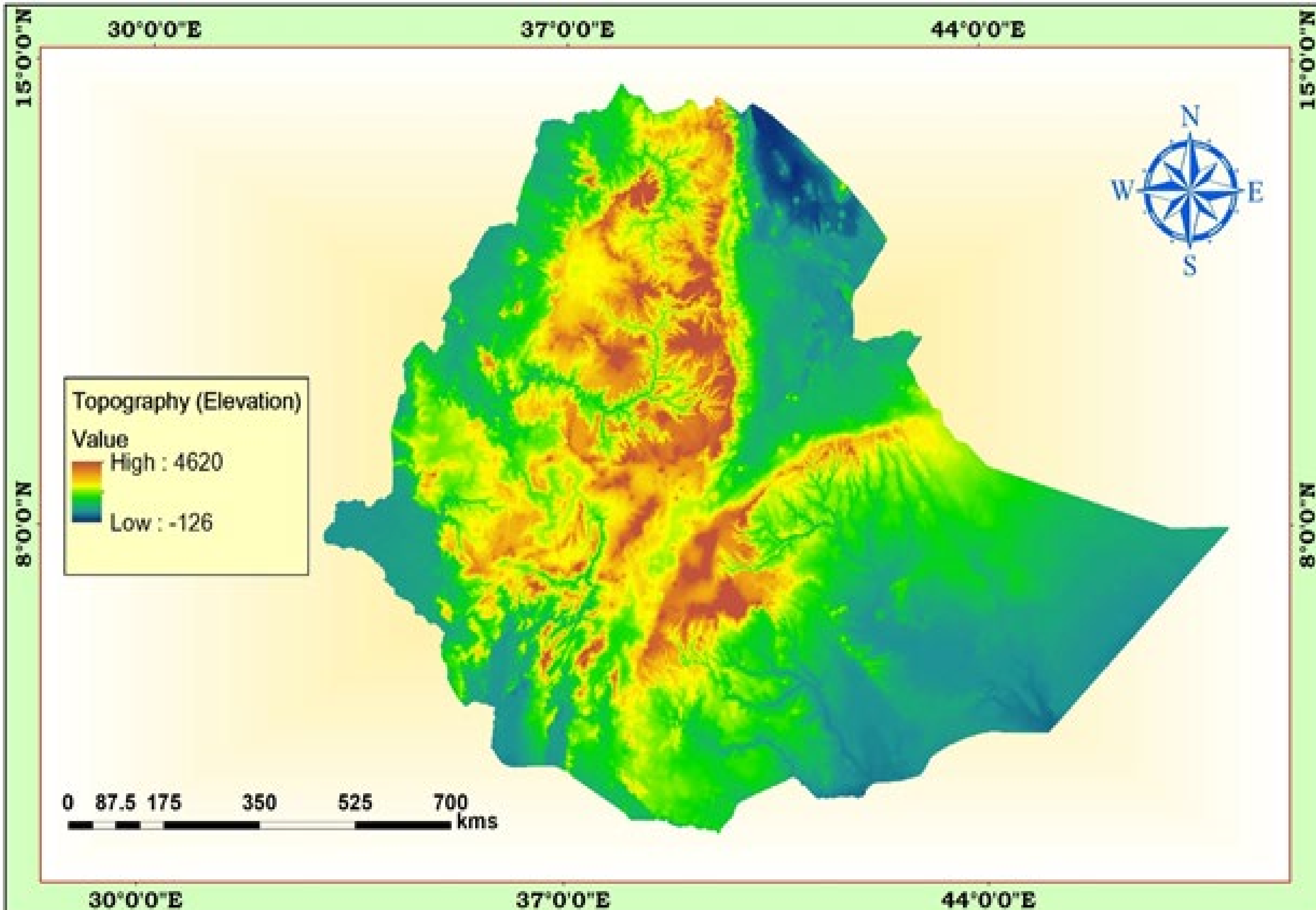
- The remaining 44% of the Ethiopian lowlands are characterized by:
  - Fewer amounts of rainfall and higher temperature.
  - High prevalence of tropical diseases.
  - Lower population densities.
  - Nomadic and semi-nomadic economic life.
  - Vast plain lands favorable for irrigation agriculture along the lower river basins.

3.2. The Physiographic Divisions of Ethiopia

- The **three major** physiographic units are:

1. The Western highlands and lowlands
2. The South-Eastern (Eastern) highlands and lowlands
3. The Rift Valley

Below is the map showing the topography of Ethiopia



3.2.1. The Western Highlands and Lowlands

- Includes all the area **west of the Rift Valley**.
- Makes up about **44%** of the area of the country.
- Encompass nearly the whole western half of Ethiopia.
- Further subdivided into **four groups of highlands** (76.3%) and **four groups of lowlands** (23.7%).

1. The Western Highlands

a. The Tigray Plateau

- Extends from the **Tekeze gorge** in the south to central **Eritrean highlands.**
- Separated from the Eritrean plateau by the **Mereb River**.
- Lies to the southeast of the upper course of the Mereb/Gash River and to the northeast of Tekeze River Gorge.
- Constitutes about **13%** of the area of the region.
- Most of the land being in between **1,000 and 2,000 meters** above sea level.

- **Denudation** has created residual features of **granite hills**, **rugged topography, and Ambas**.
- High mountains in this plateau include **Mount Tsibet** (3988 m.a.s.l), **Mount Ambalage** (3291 m.a.s.l), and **Mount Assimba** (3248 m.a.s.l).
- The famous monastery at **Debre-Damo** is also located in this plateau region.



b. North Central Massifs

- Is the **largest** in the western highlands (near to 53%).
- Much of its **northern and southern limit follows the Abay and Tekeze gorges.**
- In its **central part**, it accommodates the Lake Tana basin surrounded by plains of Fogera and Dembia in the north & an upland plain in its south.
- **58%** of the region is at an altitude of **> 2,000 meters** making it, next to the Shewan Plateau.
- Consists of the **Gonder, Wello and Gojjam Massifs**.
- Among the known mountain peaks, the most popular ones include

- \- Mount **Ras Dashen** (4,620 m.a.s.l)

  \- Mount **Weynobar/Ancua** (4462 m.a.s.l)

  \- Mount **Kidis Yared** (4453 m.a.s.l) and Mount **Bwahit** (4437 m.a.s.l) in the Simen Mountain System

  \- Mount **Guna** (4,231m.a.s.l) in the **Debre Tabour** Mountain System

  \- **Abune Yoseph** (4,260 m.a.s.l) in the **Lasta highlands of Wello** and

  \- Mount **Birhan** (4,154 m.a.s.l) in the **Choke Mountain System** in Gojjam are also part of Simen Mountain System.

- **Yeju-Wadla Delanta land bridge (ridge)** has been significant in history. It served as a route of penetration by the Turks, Portuguese, and Italians etc.

North Central Massifs…..

c. The Shewa Plateau/central highlands

- Is bounded by the Rift Valley in the east & southeast, by the Abay gorge in its northern & western limit, & the Omo gorge in the south and west.
- Occupies a **central geographical position** in Ethiopia.
- Is the **smallest** of the Western highlands (constituting **11 %**).
- Has the **largest proportion of elevated ground** (**3/4th** of its area-above 2000 m).
- Is drained, outward in all directions by the tributaries of **Abay, Omo, and Awash.**
- The tributaries of **Abay-Guder, Muger, Jema** etc. have cut deep gorges and steep sided river valleys.

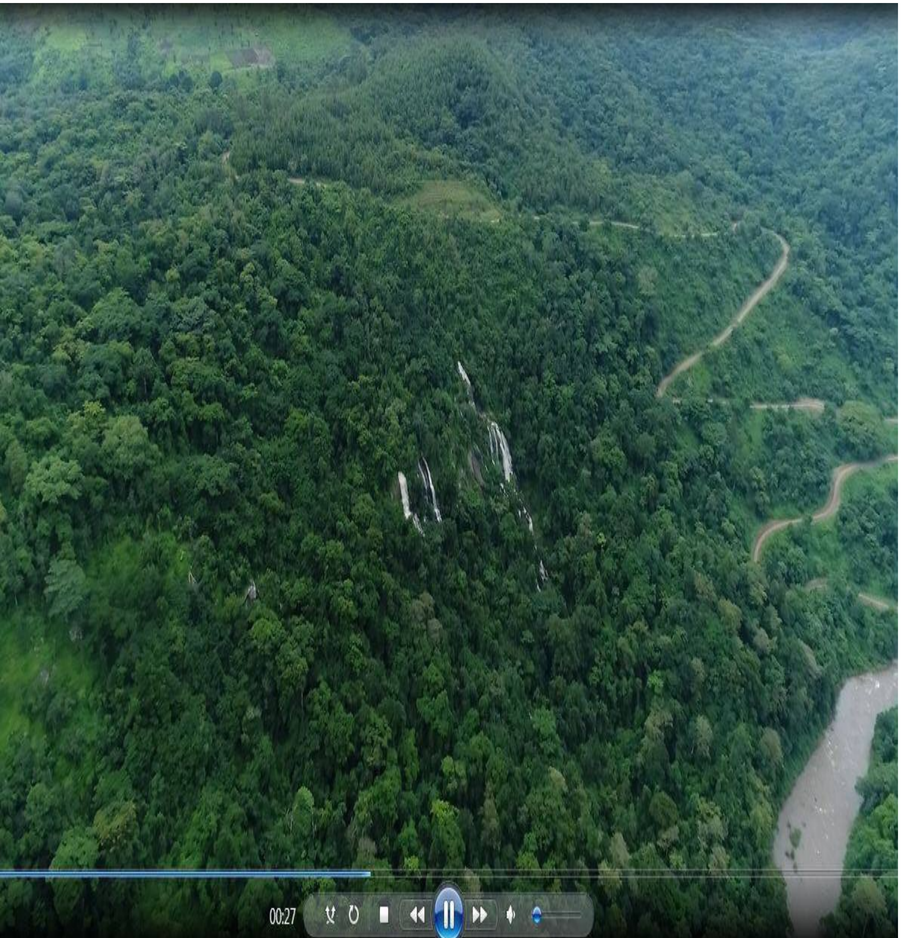
The Western Highlands and Lowlands.....

- They have created several tablelands and isolated plateau units in the north.
- The tributaries of Omo and Awash have dissected the other sides of the plateau.
- Has **relatively extensive flat-topped uplands**, giving it the **appearance of a true plateau**.
- Highest mountain in the Shewan plateau include ***Mount Abuye-Meda*** (4,000 m.a.s.l) in Northern Shewa, ***Mount Guraghe*** (3,731 m.a.s.l) in the south.

d. The South-western Highlands

- Consists of the **highlands of Wellega, Illuababora, Jimma, Kaffa, Gamo and Gofa.**
- Is separated from the adjacent highlands by the **Abay and Omo river** valleys.
- Extends from the ***Abay gorge in the north to the Kenya border and Chew Bahir in the south.***
- Accounts for **22.7%** of the area of the region & is the **second largest** in the Western highlands.
- About **70%** of its area is lies within **1,000-2,000 meters** altitude.
- Is the **wettest** in Ethiopia & is drained by Dabus, Deddessa (tributaries of Abay), Baro, Akobo and the Ghibe/Omo rivers.

- Accommodates the most **numerous and diverse ethnic linguistic groups** in Ethiopia.
- **Guge** Mountain (4,200 masl) is the highest peak in this region.



2. **The Western Lowlands**

- Are the **western foothills & border plains** that extend from **Western Tigray** in the north to **southern Gamo-Gofa** in the South.
- Make **23.7%** of the area of the physiographic region.
- The general elevation ranges between **500 and 1000 masl**.
- Is further subdivided into **four** by the protruding ridges. These are

✓**Tekeze lowland**,

✓**Abay-Dinder lowland**,

✓**Baro lowland**, and

✓**Ghibe lowland** from north to south.

- With the **exception of the Baro lowland**, the region is generally characterized by **arid or semi-arid** conditions.
- **Pastoral or semi-pastoral** economic activities dominate the area.
- The **Ghibe/Omo** lowland structurally also belongs to the **Rift Valley**.
- In the Western lowlands, there are **small but important towns** (in terms of agriculture, history, or are simply border towns and frontier ports);These are **Humera, Metema, Omedla, Kurmuk, Gambella** etc.

3.2.2. The Southeastern Highlands and Lowlands

- Is the **second largest** in terms of area & accounts for **37%** of the area of Ethiopia.
- The **highlands** make up **46%** of the physiographic division while the rest is lowland.

.

1. The Southeastern Highlands

A. The Arsi-Bale-Sidama Highlands

- Found to the east of the Lakes Region.
- Make up **28.5% of the area** of the region and **62%** of the south - Eastern Highlands.
- The well-known mountains in **Arsi Highlands** are Mount **Kaka** (4,180 m.a.s.l), Mount **Bada** (4,139 m.a.s.l) and Mount **Chilalo** (4,036 m.a.s.l).
- The **Bale highlands** are separated from the **Arsi highlands** by the head and main stream of **Wabishebelle**.
- The **Afro-Alpine** summit of Senetti plateau is found in Bale high lands.
- The highest mountain peaks in Bale Mts. are **Tulu-Demtu** (4,377 m.a.s.l) and **Mount Batu** (4,307 m.a.s.l).
- The Arsi-Bale Highlands are important **grains producing areas** with still high potential.

- The **Sidama Highlands** are separated from the Bale Highlands by the **Ghenale river** valley.
- The prominent feature in Sidama highland is the **Jemjem plateau**, an important **coffee growing area.**

- Rivers Wabishebelle and Ghenale along with their tributaries have dissected this physiographic region.
- Specially, **Weyb River**, tributary of Ghenale, has cut an underground passage (**Sof Omar cave, i.e. a spectacular cavern**) through the Mesozoic Limestone rocks.
- The cave is found near Bale Mountains.


B. The Hararghe Plateau

- Is a north-easterly extension of the south-eastern highlands & extends from the **Chercher** highland in the south-west to **Jigjiga** in the east.
- Makes up **38%** of the South Eastern highland and **17.4%** of the whole physiographic region.
- Has the **smallest proportion of upper highland** (>2,000 meters) & is a **low lying** and **elongated region**.
- The left-bank tributaries of Wabishebelle drain it.
- The highest mountain here is Mount **Gara-Muleta** (3,381 m.a.s.l).

2. The Southeastern Lowlands

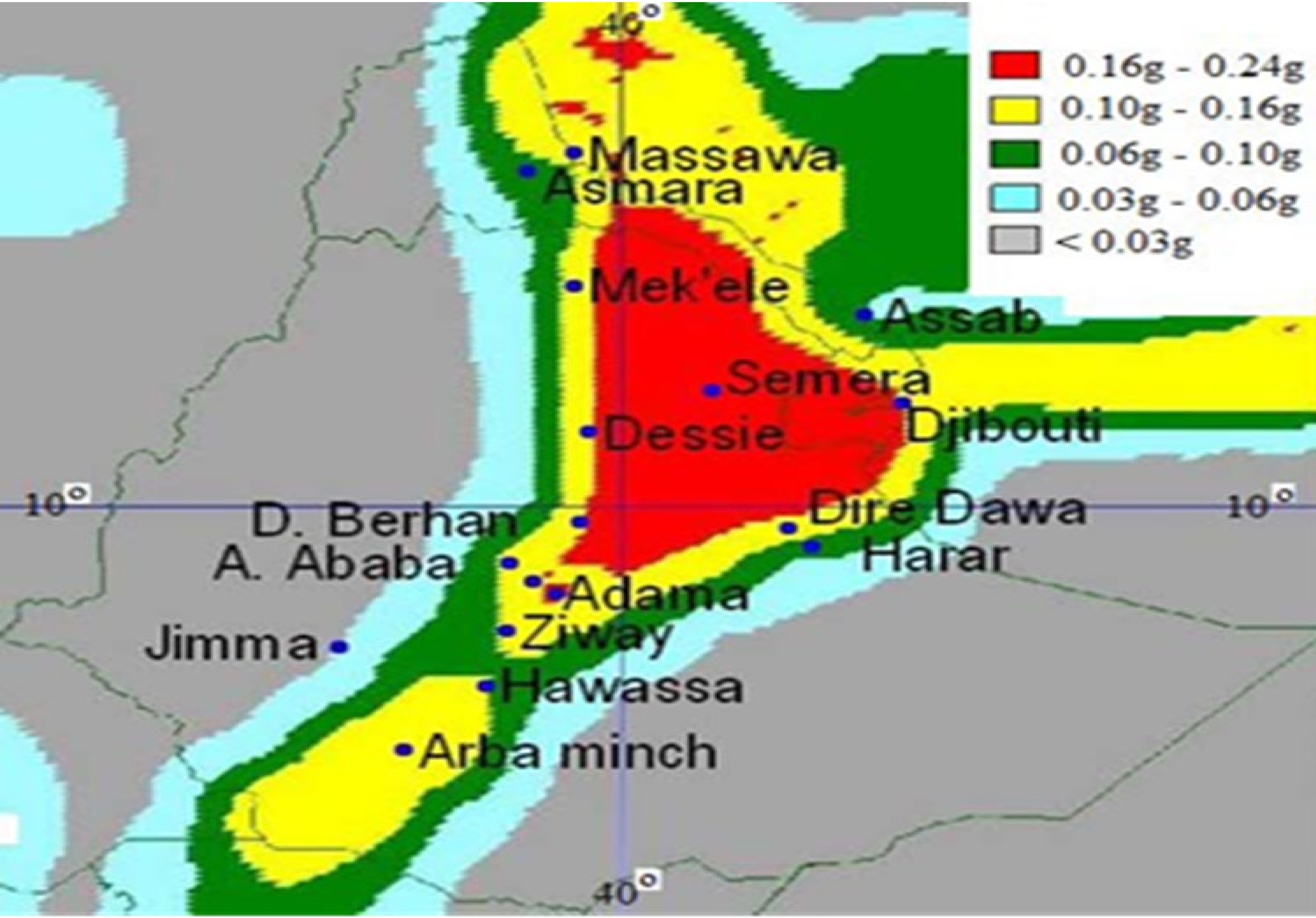
- Are located in the southeastern part of the country and are the most **extensive lowlands** in Ethiopia: Make up **54%** of the area of the physiographic region & around **one-fifth** of the country.
- Is divided into **Wabishebelle plain** (60%) & the **Ghenale** Plain (40%).
- Include the plains of **Ogaden, Elkere**, and **Borena**.
- Because of the **harsh climatic conditions**, these lowlands are **little used and support very small population.**
- Are **sparsely inhabited by pastoral and semi-pastoral communities**.
- Its economic potential includes **animal husbandry**, **irrigation**, agriculture and perhaps **exploitation of petroleum & natural gas**.

3.2.3. The Rift Valley

- Is a **tectonically** formed structural depression & is bounded by **two major** more or less **parallel escarpments**.
- Has separated the Ethiopian Highlands and Lowlands into two.
- It extends from the **Afar triangle** in the north to **Chew Bahir**
- It covers **18%** of the area of Ethiopia.
- It is **elongated and funnel shaped**, with a NE-SW orientation.
- It opens out in the **Afar Triangle**, where it is the **widest**, and **narrows down to the south.**
- The floor of the Rift Valley is made up of interconnected **troughs, grabens and depressions.**

- **Volcanic rocks**, **fluvial** and **lacustrine** deposits cover the floor.
- In many places, numerous **volcanic domes, hills and cinder cones** rise from the floor.
- Altitude in the floor ranges from **125 m.b.s.l** at **Dallol Depression**, to as high as **2,000 m.a.s.l** in the Lakes region.
- Because of its **altitudinal variation and positional differences**, the **climate also varies** from warm, hot and dry to cool and moderately moist conditions.
- Similarly, the **social and economic life reflects this pattern**.
- Sparsely inhabited by **pastoralists** where as in others parts people practice some **rain-fed agriculture**.
- It is Further subdivided into **three physiographic sub-regions**:

❖**The Afar Triangle, the Main Ethiopian Rift, and the Chew Bahir Rift**.


40°
0.16g - 0.24g
0.10g - 0.16g
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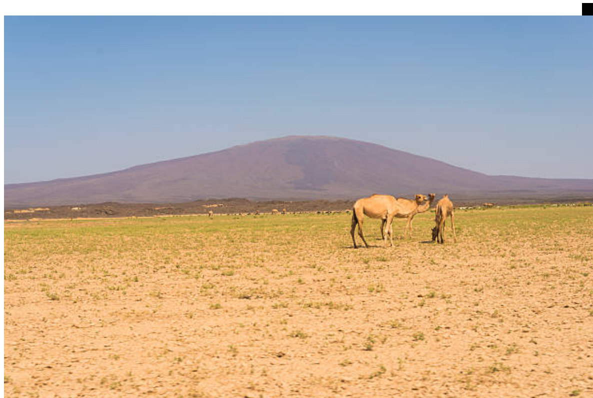
0.03g - 0.06g
< 0.03g
Massawa
Asmara
Mek'ele
Assab
Semera
Djibouti
Dessie
10°
10°
D. Berhan
Dire Dawa
A. Ababa
Harar
Adama
Ziway
Jimma
Hawassa
Arba minch
40°

i. The Afar Triangle

- Is the **largest and widest** part of the Rift Valley & makes up **54%** of the Rift Valley area.
- Is bounded by the high western and eastern escarpments in the west and east respectively, and by the Afar and Aisha Horst in the northeast.
- The area is generally of **low altitude** (300-700 meters).
- Is **triangular-shape** lowland.
- Its elevation drops from ***1,000 meters in the SW*** to ***below sea level*** in the North (Danakil depression) and in the east, where the shores of Lake Asal, fluctuating at around 125 meters below sea level
- The lowest point in this region represents the **lowest sub-aerial** point of the African continent.

- The depression, which hosts one of the most **hostile environments** on Earth (Max. temp. **> 50 Degree Celsius**, at Dallol).
- The area is characterized by **faulted depressions**, **volcanic hills, active volcanoes, volcanic ridges, lava fields and low lava platforms.**
- Lakes (**Abe, Asale, and Afrera**) occupy some of these basins.
- A prominent feature in this region is the **Denakil Depression (Kobar Sink).**
- Separated from the Red Sea by a 200 meters high land barrier, much of it lies below sea level.
- A larger part of this is covered by **thick and extensive salt plain**.
- Lake Asale and Lake Afrera occupy the lowest parts of this sunken depression.
- The Afar Triangle is generally **hot and dry**.
- The only respite one gets in the Southern part is from the waters of the **Awash River**.

The **economic importance** of this region includes **salt extraction**, **irrigation** along the Awash River & **electric potential** from **geothermal energy**.



ii. The Main Ethiopian Rift/Central Rift

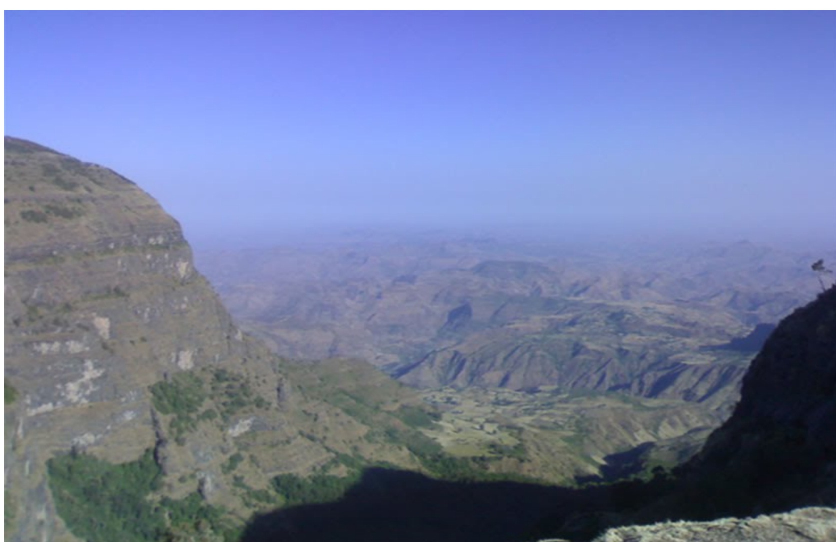
- Refers to the **narrow belt of the Rift Valley** that extends from **Awash River** in the north to **Lake Chamo** in the south.
- With the **exception of the Arbaminch area**, the bounding **escarpments are generally low**.
- Is the **narrowest & the highest** & has an average width of **50-80 kms** and general elevation of **1,000-2,000** m.a.s.l.
- The floor in many places is dotted by **cinder cones & volcanic mountains.**
- The big ones include Mount **Fentale, Boseti-guda** (near Adama), **Aletu** (north of Lake Ziway) & **Chebi** (north of Lake Hawasa).

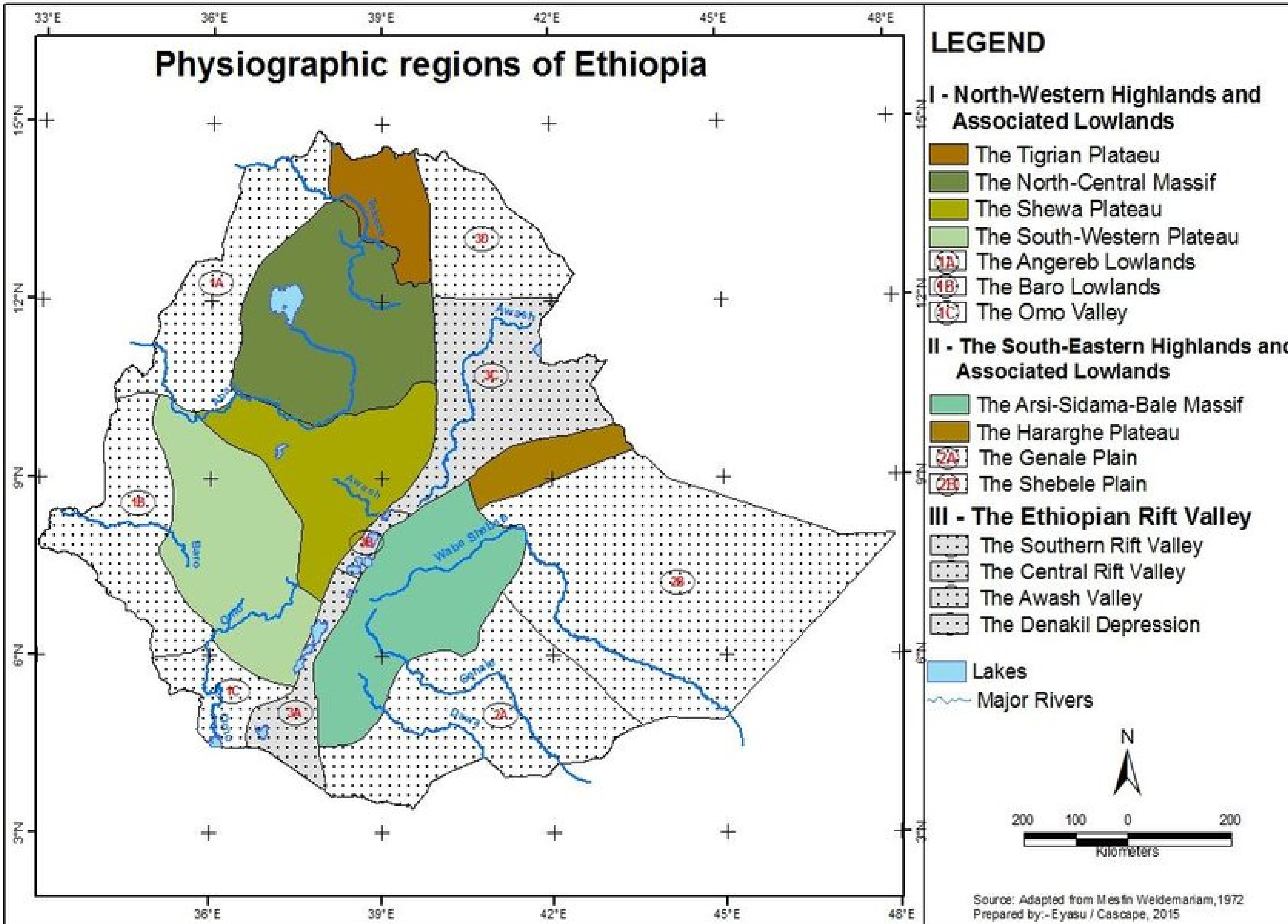
- Contains **numerous lakes** formed on **tectonic sags and fault depressions.**
- Is generally **milder and watery**.
- Here **rain-fed agriculture** is practiced.
- Other resource bases include the **recreational value of the lakes**, the **agricultural importance** of some streams and lakes, and the **geothermal energy potential**.


Getty Images


iii. The Chew Bahir Rift

- Is the **smallest** and the **southern-most part** of the Rift Valley.
- Gneissic highlands of **Konso** and the surrounding highlands separate it from the Main Ethiopian Rift to the north.
- The characteristic feature of this region is the **broad and shallow depression**, which is a **marshy area** covered by tall grass, into which the Segen and Woito streams empty.





Physiographic regions of Ethiopia
LEGEND
I - North-Western Highlands and Associated Lowlands
The Tigrian Plataeu
The North-Central Massif
The Shewa Plateau
The South-Western Plateau
1A The Angereb Lowlands
1B The Baro Lowlands
1C The Omo Valley
II - The South-Eastern Highlands and Associated Lowlands
The Arsi-Sidama-Bale Massif
The Hararghe Plateau
2A The Genale Plain
2B The Shebele Plain
III - The Ethiopian Rift Valley
The Southern Rift Valley
The Central Rift Valley
The Awash Valley
The Denakil Depression
Lakes
Major Rivers
N
200 100 0 200
Kilometers
Source: Adapted from Mesfin Weldemariam,1972
Prepared by:- Eyasu / Cascape, 2015
33°E 36°E 39°E 42°E 45°E 48°E
15°N 12°N 9°N 6°N 3°N
Awash
Baro
Omo
Wabe Shebele
Genale
Dawa

3.3. The Impacts of Relief on Biophysical and Socioeconomic Conditions

- The **highly dissected character** of the landscape over much of the country influence the various **socioeconomic aspects** of Ethiopia as presented hereunder.

1. **Agricultural practices**

- Relief influences **farm size and shape** in that in an area of rugged terrain the farmlands are **small in size and fragmented** and tend to be **irregular in shape**.

❖ Choice of **farming techniques** and **farm implements** are highly influenced by relief.

❖ Relief influences **crop production**.

❖ The practice of **animal husbandry** is also influenced by relief.

2. **Settlement pattern**

- Rugged & difficult terrain hinders the **development of settlement and its expansion.**
- Highlands of Ethiopia are **densely settled**.

- The **highlands** of Ethiopia are characterized by **sedentary life** & **permanent settlements** while **lowlands** that are inhabited by **pastoralists** have **temporary settlements**.

3. Transportation and communication

- The highly dissected nature of the landscape is a **barrier** to the development of transportation.
- The difficult terrain makes infrastructure development & maintenance **costly**.
- **TV and radio** communications are also highly influenced by relief.
- The rugged topography rendered rivers **less navigable** due to the **waterfalls, deep gorges and steep cliffs**.

4. Hydroelectric power potential

- The great **difference in altitude** coupled with **high rainfall** created **suitable conditions** for a very high potential for the production of HEP in Ethiopia.

5. Socio-cultural feeling

- The rugged terrain as a result of excessive surface dissection resulted in **isolation of communities** that led to the occurrence of **cultural diversity.**
- People who live in the highlands - ***degegnas*** **(mountaineers**) and those who live in the lowlands-***kollegnas*** **(lowlanders**).

6. Impacts on climate

- The **climate of Ethiopia** is a result of the **tropical position** & the **great altitudinal variation**.
- **Highlands** with **higher amount of RF & lower rate of Evaporation** tend to be **moisture surplus** compared to the moisture deficit lowlands.

7. Impacts on soil

- **Steep mountain slopes** have **shallow and little developed soils**.

8. Impacts on natural vegetation

- Relief through its effect on climate and hydrology affect the type of **natural vegetation** grown in an area.

Chapter Four
DRAINAGE SYSTEMS AND WATER RESOURCE OF ETHIOPIA AND THE HORN

4.1 Introduction

- About **71%** of the earth's total surface is covered by water bodies majorly occupied by **seas and oceans**.
- Of the earth's total water surface, nearly **97.5% is alkaline** accumulated in seas and oceans.
- **Alkaline water** is water that is **slightly basic**. It contains basic minerals such as **calcium, magnesium, or bicarbonate**. These compounds bind to hydrogen ions in solution, making the water more basic.
- The remaining **2.5% is fresh water**, of which nearly **68.7% is deposited in glaciers**, **30.1% in ground water**, **0.8% in permafrost and 0.4% in surface waters.**
- Water in lakes, rivers, atmosphere, soils and wetlands are considered as **surface waters**.
- Surface and ground waters are by far **the most abundant and easily available** fresh waters.
- However, fresh water is distributed **unevenly** throughout the world following varied **latitudinal locations, climatic and topographic setups**.

- As you have discussed in the third chapter, the topographic setup of Ethiopia is characterized by **complex blend of massive highlands, rugged terrain, and low plains.**
- The **diverse topographical setup**, relatively **higher rainfall** and its nearness to equator made the country to have **larger volume of ground and surface water.**
- Around **0.7%** of the total land mass of Ethiopia is covered by water bodies.
- Although it requires further detailed investigation, the country's surface water potential as studied in different integrated river basin master plans is estimated to be **124.4 billion cubic meters** (BCM).
- Consequently, many call Ethiopia, **the water tower of *"Eastern Africa"*.**

4.2. Major Drainage System of Ethiopia

- The flow of water through well-defined channel is known as **drainage**.
- **A drainage system** is made up of a **principal river and its tributaries** (the rivers that flow into it).
- A river system **begins** at a place called the **source** or **headwater** and **ends** at a point called **mouth**.
- Therefore, a drainage system is **branched network of stream channels** together with the adjacent land slopes they drain.
- The drainage pattern of an area is the **outcome of the geological processes, nature and structure of rocks, topography, slope, amount and the periodicity of the flow.**
- A drainage basin is the topographic region from which **a river and its tributaries collect both the surface runoff and subsurface flow**.
- It is bounded by and separated from other river basins by **a water divide** or ***topographic divide***.

Major Drainage System of Ethiopia……

- *The general patterns of major river basins in Ethiopia are determined by* ***topographical structures*** *which can be clarified as:*

a. The topography of the **outward sloping** of the ***Western and South eastern plateaus***

b. The structural formation ***of the Rift Valley with its in-ward-sloping escarpments resulting mainly in an inland drainage system***.

c. **Faults and joints** that structurally influence part of the courses of many rivers.

The Major Drainage Systems

- Following the complex physiographic setup and geological makeup, Ethiopia possesses **three** broadly classified drainage systems namely ***Western, Southeastern and Rift Valley Drainage Systems.***

- *Western and the Southeastern drainage systems* ***are separated by the Rift Valley system.***

4.2.1. The Western Drainage Systems

- The Western Drainage Systems are **the largest** of all drainage systems **draining 40 percent** of the total area of the country and **carry 60 percent of the annual water flow**.
- Most of the catchment area coextends with the westward sloping part of the western highlands and western lowlands.
- This drainage system comprises **four major river basins** namely the **Tekeze, Abay, Baro-Akobo, Ghibe (Omo).**
- Unlike other river basins in the system, the **Ghibe (Omo) flows southward**.
- The Abay, Tekeze and Baro **flow westward** ultimately joining the **Nile** which finally ends at Mediterranean Sea.

- The **largest river** both in **volumetric discharge and coverage** in the western drainage systems is the ***Abay.***
- *Abay river basin covers an area of **199,812 km2**, covering parts of Amhara, Oromia and Benishangul-Gumuz regional states.*
- *Together with its tributaries, most of which are* left-bank tributaries; it carries ***65 percent*** *of the annual water flow of the region.*
- *Abay which rises from Lake Tana (some sources indicate its origin from Sekela, Choke mountain)* ***flows about 1,450 kilometers*** *and joins the White Nile in Khartoum, Sudan to form the Nile River.*

The Western Drainage Systems…..

- *More than **60 streams** drain the Abay within elevation ranging between 500 - 4261 meters above sea level.*
- *The largest of these is **Ghilgel Abay** (Little Abay).*
- *Abay flows eastward, turns 180 degree to make a large bend and after cutting an impressive and deep gorge emerges out in the west.*
- Similarly, the ***Tekeze and its tributaries**, carrying **12 percent** of the annual water flow of the region drains **82,350 Km2** of land surface within elevation ranging between 536-4517 meters above sea level.*
- *Erosion in the basin resulted in large tablelands, plateau blocks and isolated mountain groups.*

The Western Drainage Systems…..

- *The basin has two main tributaries (**Angereb and Goang**) which rises in the central highlands of Ethiopia.*
- *Tekeze River is termed **Atbara in Sudan**, which is a tributary of the Nile.*
- *The total mean annual flow from the basin is estimated to be **8.2 billion** metric cubes (BMC).*
- **The *Baro-Akobo and Ghibe / Omo rivers*** *drain the **wettest** highlands in the south and southwestern Ethiopia.*
- *They carry **17 percent** and **6 percent** of the annual water flow respectively.*
- *The Ghibe/ Omo river basin drains an area of **79,000 km2** with an estimated mean annual flow of **16.6 BMC**.*

- *In the lower course, the Baro River flows across an extensive marshy land.*
- *Baro/Akobo river basin has an area of* ***75,912 km2****, covering parts of the Benishangul-Gumuz, Gambella, Oromia, and SNNPR.*
- *The total mean annual flow from the river basin is estimated to be* ***23.6 BMC.***
- *The Baro together with Akobo forms the* ***Sobat River in South Sudan****.*
- *The Ghibe / Omo River finally empties in to the* ***Chew-Bahir*** *at the mouth of* ***Lake Turkana*** *(an elongated Rift Valley lake) thereby forming an inland drainage.*

4.2.2. The Southeastern Drainage Systems

- Nearly the entire physiographic region of southeastern part of Ethiopia is drained by the southeastern drainage systems.

- The basin which is mainly drained by ***Wabishebelle*** *and* **Ghenale,** *slopes south-eastwards across large water deficient plains*.
- *Major highlands of this basin include plateaus of Arsi, Bale, Sidama and Harerghe.*
- *Wabshebelle and Ghenale rivers cross the border into Somalia, carrying **25 percent** of the annual water flow of Ethiopia.*
- Ghenale River basin has an area of **171,042 km2**, covering parts of Oromia, SNNPR, and Somali regions.
- Ghenale, which has **fewer tributaries** but **carries more water than** *Wabishebelle, reaches* **the Indian Ocean**.

- The Ghenale basin flows estimated to be **5.8 BMC** within elevation ranging between 171-4385 meters above sea level.
- In Somalia it is named the ***Juba River***.
- **Wabishebelle** with a total catchment area of **202,697 km2**, is the **largest river in terms catchment area**.
- It drains parts of Oromia, Harari and the Somali regions.
- It is the **longest river in Ethiopia**.
- Its tributaries are mainly **left bank and, most of them, are intermittent**.

The Southeastern Drainage Systems .....

- Despite its size, the Wabishebelle **fails to reach the Indian Ocean** where at the end of its journey it flows parallel to the coast before its water disappears in the sands, **just near the *Juba River***.

4.2.3. The Rift Valley Drainage System

- The Rift Valley drainage system is **an area of small amount of rainfall**, **high evaporation** and **small catchment area**.
- The size of the drainage area is restricted by the outward sloping highlands, which starts right from the edge of the escarpment.
- The Rift Valley drainage system is therefore left with the **slopes of the escarpment** and the **Rift Valley floor** itself as the catchment area.
- The only major river basin is that of the **Awash**.
- Awash river basin has a catchment area of **114,123 km2** and has an average annual discharge of **4.9 billion cubic meters**.

- The Awash River originates from ***Shewan plateau*** *in Central highlands of Ethiopia, and flows 1250 kms.*
- *It covers parts of the Amhara, Oromia, Afar, Somali, Dire Dawa, and Addis Ababa City Administration.*
- *Awash is the* ***most utilized river*** *in the country.*
- In the Rift Valley drainage systems, there is **no one general flow direction, as the streams flow in all directions.**
- Following the rift valley orientation the Awash flows in a **NE direction.**
- It finally ends in a maze of small lakes and marshy area; the largest of which is **Lake Abe** on the Ethio-Djibouti border.

- The **Afar drainage sub-basin** has practically **no stream flow**.
- It is an area of **little rain**, **very high temperature** and **very high evaporation.**
- *Lake **Afrera** and **Asale** are the only main surface waters in the basin which are not the result of any meaningful surface flow.*
- *Their formation is related to tectonic activities*.
- The **Southern part** of the Rift Valley sub-basin is characterized by a **number of lakes** and **small streams**.
- It is also described as **lakes region**.
- The lakes occupy fault depression

- There are small streams that drain down from the nearby mountain slopes which supply water to the lakes.
- For example,

✓*Meki and Katar Rivers flow into Ziway*

✓*Bilate into Abaya and*

✓*Segen into Chew Bahir.*

- *Likewise, some of these lakes are interconnected.*
- *Lakes Ziway and Langano drain into Lake Abijiata through the small streams of Bulbula and Horocolo respectively.*

4.3. Water Resources: Rivers, Lakes and Sub-Surface Water

4.3.1. The Ethiopian Rivers

- Unlike many other African countries, Ethiopia is endowed with **many rivers.**
- Majority of the rivers originate from **highland areas** and **cross the Ethiopian boundary.**
- Altogether, Ethiopian rivers form **12 major watersheds** separating the Mediterranean Sea from the Indian Ocean drainage systems.

Table 4.1: Data on major Ethiopian rivers

River	Catchment Area(km^2)	Annual Volume BMC	Terminus/ Mouth	Major tributaries
Abay	199,812	**54.5**	Mediterranean	Dabus, Dedessa, Fincha, Guder, Muger, Jema, Beshilo
Wabishebelle	**202,697**	**3.4**	Coast of Indian Ocean	Ramis Erer, Daketa Fafan
Genale Dawa	171,042	6	Indian-Ocean	Dawa, Weyb, Welmel, Mena
Awash	114,123	4.9	Inland-within Ethiopia	Akaki, Kesem, Borkena, Mile
Tekeze	87,733	8.2	Mediterranean	Goang, Angereb
Gibe (Omo)	79,000	16.6	Lake Turkana	Gojeb
Baro Akobo	**75,912**	23.23	Mediterranean	Akobo

- Owing to the **highland nature** of the Ethiopian landmass, **surface ruggedness**, the **outward inclination of the highlands**, and the **climatic conditions**, Ethiopian rivers have the following characteristics.

❖ Almost all major rivers **originate from the highlands** elevating more than 1500 meters above sea level,

❖ Majority of Ethiopian rivers are **trans-boundary**,

❖ Due to the marked seasonality of rainfall, Ethiopian rivers are characterized by **extreme seasonal fluctuation**.

❖ In the **wet season**, **runoff is higher** while during the **dry seasons** they became mere **trickles of water/even dry up**,

- Due to surface ruggedness they have **rapids and waterfalls** along their course,
- They have **cuts, steep-sided river valleys** and **deep gorges** along their courses,
- Rivers in Ethiopia flow on **steep slopes** having steep profiles.
- Some of the rivers *serve as **boundaries**, both international and domestic administrative units*.

4.3.2. The Ethiopian Lakes

- Relatively Ethiopia is **rich in lakes**.
- Almost all Ethiopian lakes are result of **tectonic process** that took place during ***Quaternary period of Cenozoic era***.
- *Except few Ethiopian lakes, majority of lakes are located within the Rift Valley System.*
- *The lakes in the drainage are mainly formed on faulted depressions and are clustered along the system forming linear pattern.*
- Lake *Tana, the largest lake in Ethiopia* ***occupies a shallow depression in the highlands.***

- *The Tana depression is believed to be formed following **slower sinking** and **reservoir by lava** flow between Gojjam and Gonder massifs.*
- *Ethiopia is also gifted with **crater lakes**.*
- *These include the lakes at and around Bishoftu, Wonchi (near Ambo), Hayk (near Dessie) and the Crater Lake on top of Mount Zikwala.*
- *Lake Ashenge (Tigray) is formed on a tectonic basin.*
- *Other types of lakes in Ethiopia are **man-made** such as Lakes Koka, Fincha and Melka Wakena, and many other lakes dammed following hydroelectric power generation projects.*

The Ethiopian Lakes ……

- Cluster of lakes are lined up within main Ethiopian rift.
- ***Lake Abaya*** *is the largest of all the lakes in the system.*
- *The southern tip of the Rift Valley forms the marshy land called the* ***Chew Bahir*** *which is drained by Segan and Woito.*
- *Shala and Ziway are the* ***deepest and the shallowest*** *lakes in the central Ethiopian Rift*

Table 4.2: Area and depth of some of Ethiopian Lakes

Lake	**Area (km^2)**	**Max. Depth(m)**	**Lake**	**Area (km^2)**	**Max. Depth(m)**
Tana	**3600**	9	Abijata	205	14
Abaya	1162	13.1	Awassa	129	10
Chamo	551	13	Ashenge	20	25
Ziway	442	**8.95**	Hayk	**5**	23
Shala	409	**266**	Beseka	48.5	11
Koka	205	9			

4.3.3. Subsurface (Ground) Water Resource of Ethiopia

- As compared to surface water resources**, Ethiopia has lower ground water potential.**
- However, there exists higher total exploitable groundwater potential.
- **Climatic** and **geophysical conditions** determine the availability of groundwater resource.
- Based on existing scanty knowledge, the groundwater potential of Ethiopia is estimated to be ***2.6 - 6.5 BMC***.
- *However, this estimate is now considered **underestimated**.*
- *Considering various separate studies, Ethiopian potential of groundwater is believed to range **between 12-30 BMC**.*

4.4. Water Resources Potentials and Development in Ethiopia

- The enormous water resource potential of Ethiopia is **underutilized** due to so many factors.
- However, there are plenteous of opportunities that can transform the resource into our collective social and economic needs.
- The followings are **some of potential development uses** of water resource of Ethiopia.

a) Hydro-electric Potential

- Ethiopian rivers have a very high potential for generating electricity.
- The exploitable potential of hydroelectric power is estimated at about **45000 megawatts**.

- The **first hydroelectric power** generation plant was installed on **Akaki River (Aba Samuel) in 1932.**
- Currently many hydroelectric power dams are operating and many others are under construction to realize Ethiopia's ambitious energy goals.
- **Grand Ethiopian Renaissance Dam** (GERD) is the country's largest dam under construction aiming to generate **6400 megawatts**.
- **Gilgel Gibe III** hydropower project has gone operational generating **1870 megawatts**.

- Currently Ethiopia is administering **14 hydroelectric power plants** constructed on Lake Aba Samuel, Koka, Tis Abay, Awash, Melka Wakena, Sor, Fincha, Gibe/Omo, Tana Beles and Tekeze, generating close to **4000 megawatts of energy**.
- Besides the domestic use of generated electricity, the country is **exporting electricity** to the neighboring countries.
- The **major problem** related to the use of Ethiopian rivers for the generation of hydroelectric power is the **seasonal flow fluctuations and impact of climate change and variability.**
- The **severe erosion from the highlands and sedimentation** in the reservoirs is also a critical problem for hydroelectric power generation.

b) Irrigation and Transportation

- The terrain in Ethiopia is so **rugged** that **it limits the uses** of Ethiopian rivers both for irrigation and transportation.
- In the **highlands**, steep slopes, rapids, waterfalls, narrow and deep valleys and gorges are important obstacles.
- But on the **lowlands**, their demand for irrigation is high.
- Regardless of existing physiographic setups, Ethiopia's **potential of irrigation** is estimated to be **5.3 million hectares**.
- The *Baro-Akobo and Genale Dawa river systems have **large irrigation** potential compared to other basins.*

- *Despite the untapped irrigation practice, more than **60%** of the area under irrigation so far is located in **Rift Valley Drainage System**.*
- *Except few, **majority of hydro-electric reservoirs** are **multi-purpose** and are expected to contribute for irrigation.*
- Majority of Ethiopian rivers **are not suitable for transportation**.
- The ***Baro*** *at its lower course is the only navigable river.*
- *Comparatively, **Ethiopian lakes** are much suitable for transportation than rivers.*
- ***Lake Tana*** *and* ***Abaya*** *are relatively the most used for transportation*

c) Fishing and Recreation

- The majority of Ethiopian lakes are **rich in fish**.
- Currently the annual production of fish is estimated to be **31.5 thousand tons.**
- The exploitable **potential** is however, by far greater than the **current production.**
- **Exploitable fish potential in lakes varies**.
- Currently **Lake Tana leads the potential** by estimated 8,000-10,000 tons per year.
- Fish production from **Lake Chamo** is estimated at 4,500 tons per year.

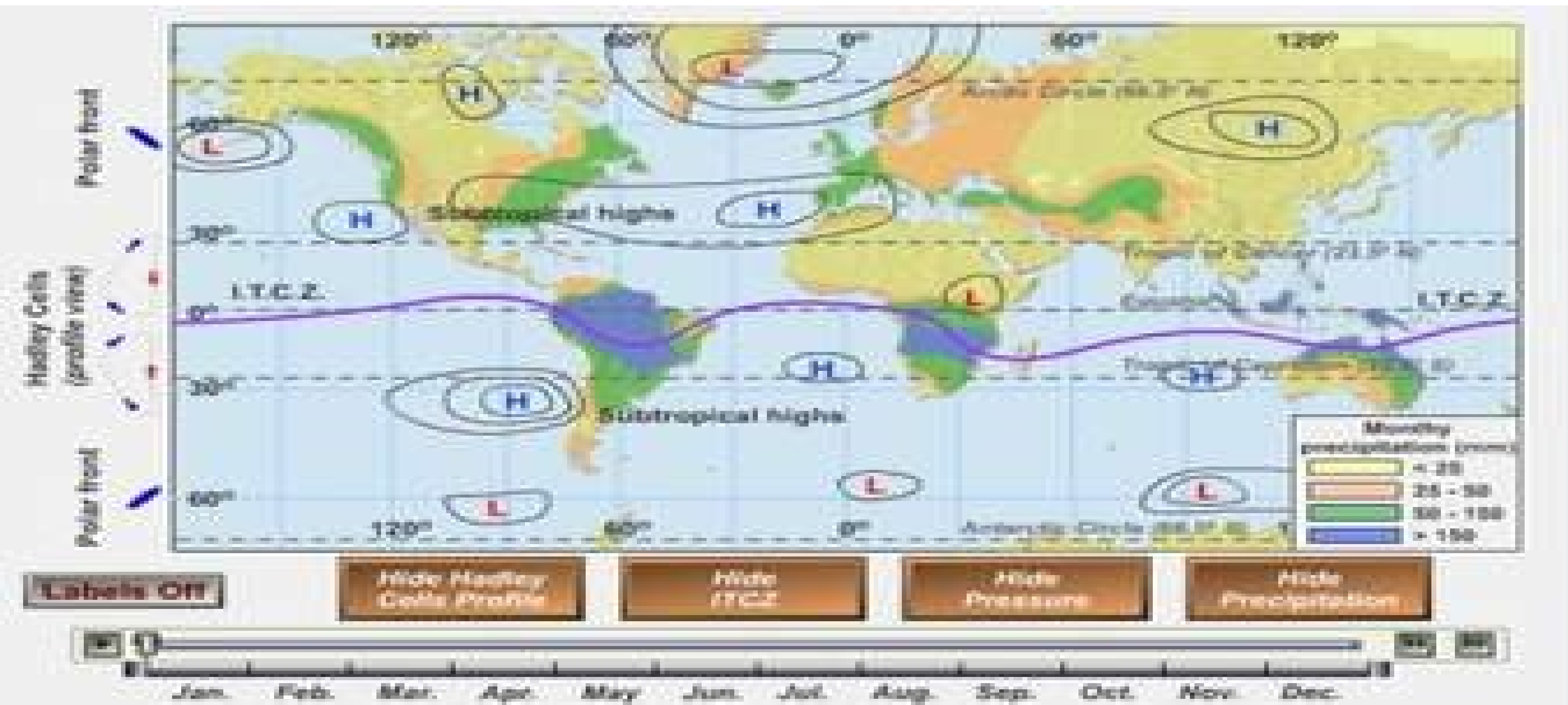
- However, more than **60% of fish supplies** are coming from Ethiopian main Rift Valley lakes.
- However, some of the lakes are currently threatened by **sedimentation**, **invasive species**, **over exploitation** and **expansion of investments around lakes.**
- There are a **variety of fish, birds** and **other aquatic life** forms in the lakes, some of which are only **endemic to Ethiopia**, with immense scientific purposes.
- This and the scenic beauty of the lakes, the hot springs, the spectacular river gorges & the most impressive waterfalls make Ethiopian rivers & lakes **important recreational & tourist attractions**.

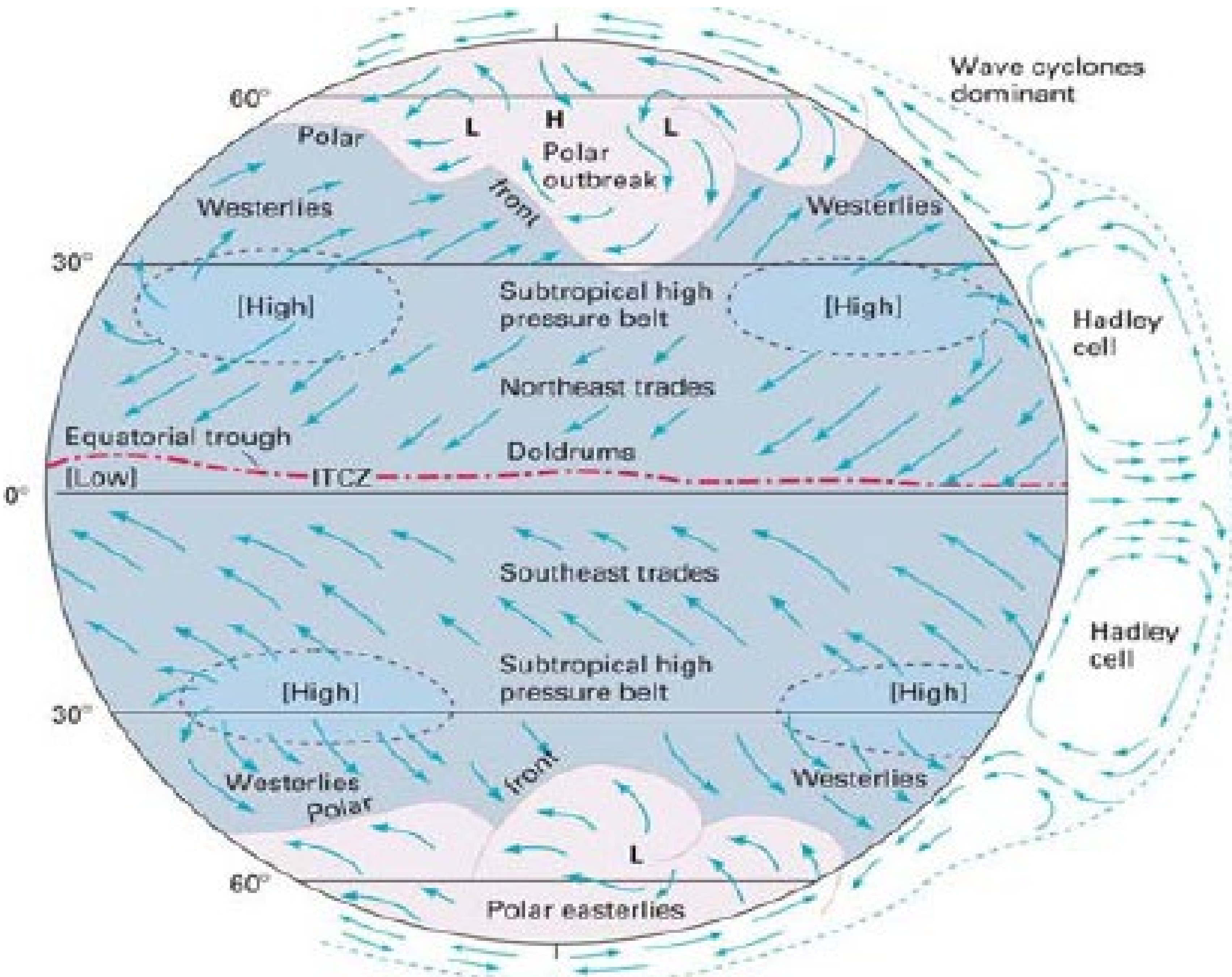
Chapter 5 : The Climate Of Ethiopia & the Horn Of Africa

5.1 Introduction

- Ethiopia, is characterized by a wide **variety of altitudinal ranges and diverse climatic conditions.**
- In addition, because of its **closeness to the equator** and **the Indian Ocean**, the country is subjected to large **temporal and spatial variations** in elements of weather and climate.
- The climate of Ethiopia is, therefore, mainly controlled by the **seasonal migration of the *Intertropical Convergence Zone*** (ITCZ) and associated **atmospheric circulations** as well as by its **complex topography.**
- The convergence of **Northeast Trade winds** forms the ITCZ, which is a **low-pressure zone**.

- ITCZ, circles the Earth generally near the equator where the trade winds of the Northern and Southern Hemispheres come together.
- It is characterized by convective activity which generates often vigorous thunderstorms over large areas.
- It is most active over continental land masses by day and relatively less active over the oceans.

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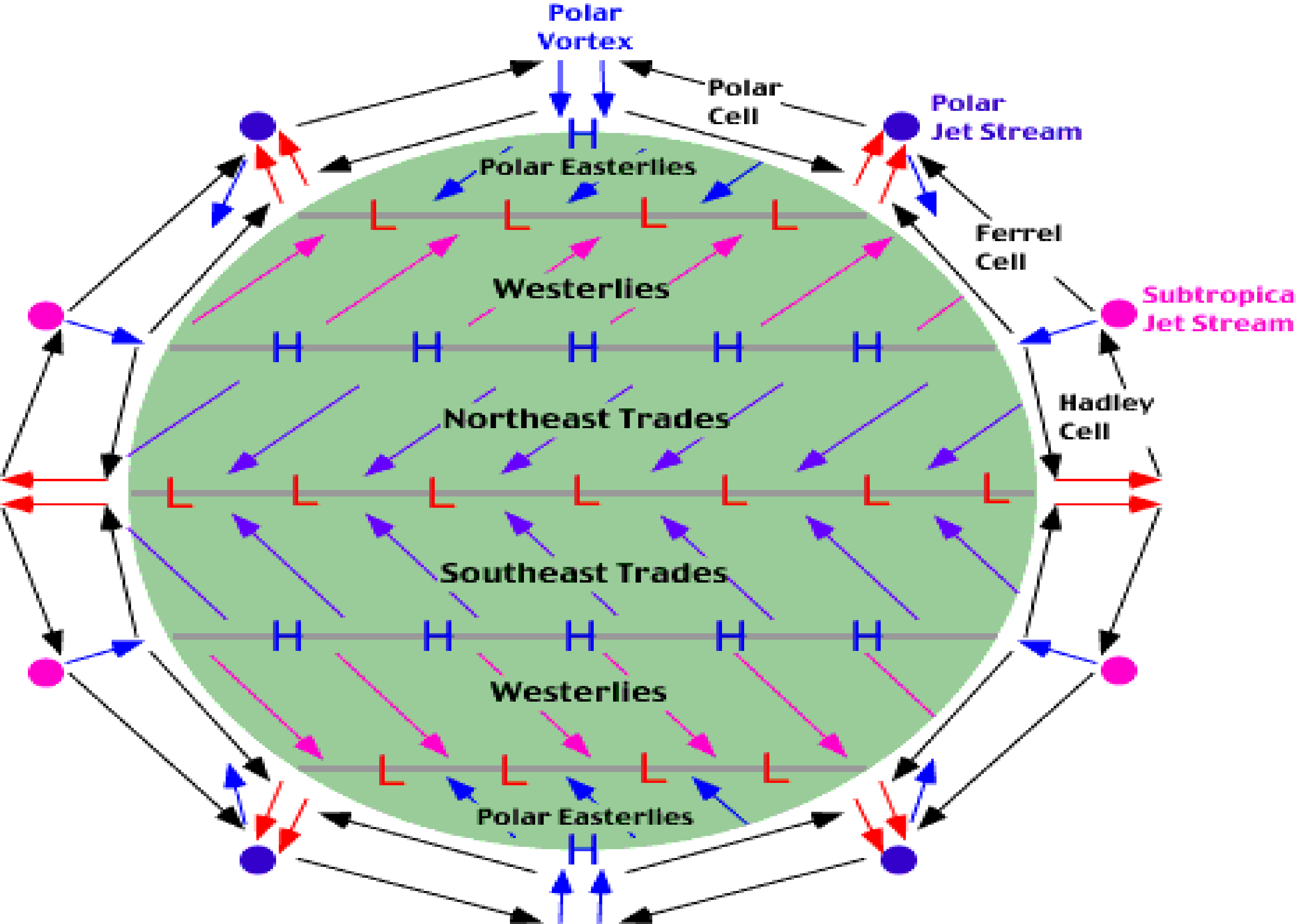
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Vortex
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Jet Stream
Polar Easterlies
Ferrel
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Westerlies
Subtropical
Jet Stream
Hadley
Cell
Northeast Trades
Southeast Trades
Westerlies
Polar Easterlies

- **Weather** is the **instantaneous** or current state of the atmosphere composing temperature, atmospheric pressure, humidity, wind speed and direction, cloudiness and precipitation.
- **Climate** refers the state of the atmosphere over long periods, decades and more.
- It is the composite of **daily** weather conditions recorded for long periods of time.

5.2. Elements and Controls of Weather and Climate

- All weather conditions may be traced to the **effect of the Sun** on the Earth.
- Most changes in weather involve large scale horizontal motion of air which is called **wind.**
- Here are lists of major elements and controls of weather and climate.

Table 5.1 Elements and controls of weather and climate

Elements	**Controls**
1. Temperature	1 Latitude/angle of the Sun
2. Precipitation and humidity	2 Land and water distribution
3. Winds and air pressure	3 Winds and air pressure
	4 Altitude and mountain barriers
	5 Ocean currents

5.2.1. Controls of Weather and Climate

- What do you think is the source of **summer rainfall in Ethiopia**?
- Have you ever noticed varying lengths of days and nights by seasons? What do you think is the reason behind?
- The climate of any particular location on earth is determined by a **combination of many interacting factors**.
- These include **latitude, elevation, nearby water, ocean currents, topography, vegetation, and prevailing winds**.
- Moreover, the **global climate system** and any changes that occur within it also influence **local climate**.

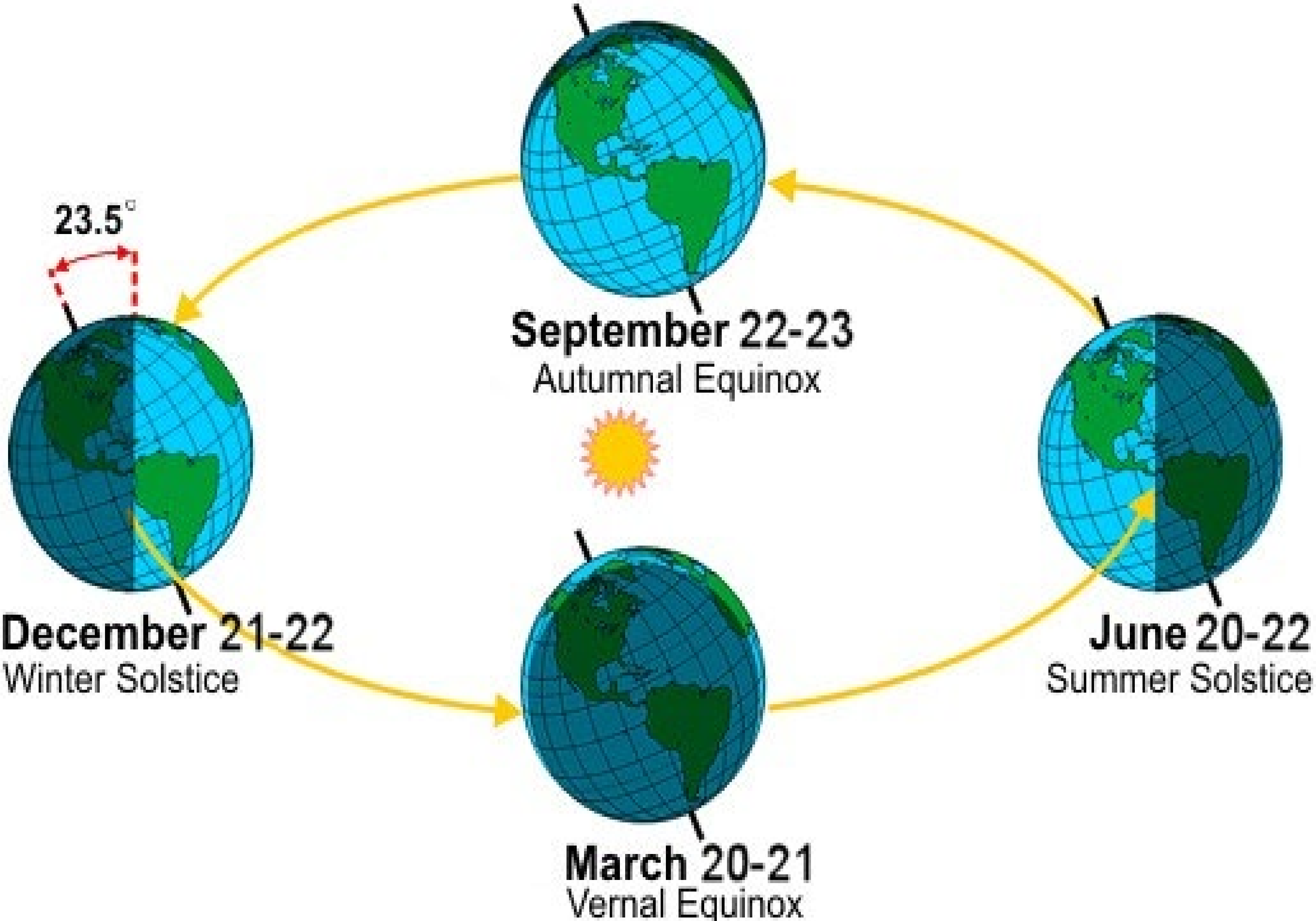
a. Latitude

- Latitude is the **distance of a location from the equator** measured in degrees.
- The sun shines directly on **equator** for more hours during the year than anywhere else.
- As you move further away from the equator towards the poles, less solar insolation is received during the year and the temperature become colder.
- Ethiopia's latitudinal location has bearings on its temperature.
- **Latitudinal location of Ethiopia** and the Horn resulted in:

  - High average temperatures,
  - High daily and small annual ranges of temperature,
  - No significant variation in length of day and night between summer and winter.

b. Inclination of the Earth's Axis

- The Earth's rotation axis makes an angle of about **66½°** with the plane of its orbit around the sun, or about **23½°** from the perpendicular to the ecliptic plane.
- This inclination determines the location of the Tropics of Cancer, Capricorn, and the Arctic and Antarctic Circles.
- As the earth revolves around the sun, this inclination **produces a change in the directness of the sun's rays**; which in turn causes the **directness of the sun** and **differences in length of day and seasons**.


23.5°
September 22-23
Autumnal Equinox
December 21-22
Winter Solstice
June 20-22
Summer Solstice
March 20-21
Vernal Equinox

Equinoxes and Solstices

- An **equinox** is the instant of time when the sun strikes the plane of the Earth's equator.
- During this passage the length of **day and night are equal**.
- Moreover, revolution of the earth along its orbit, the inclination of its axis from the plane of that orbit, and the constant position (parallelism) of the axis causes seasonal changes in the daylight and darkness periods.
- Equinox appears **twice a year**.
- There are two types of equinoxes:

1. The Vernal (Spring) equinox

- It is the day when the point of verticality of sun's rays crosses the equator **northwards**.
- It experiences in Northern Hemisphere when the sun is **exactly above the equator.**
- During this period, the length of day and night are equal.
- It marks the beginning of spring season.
- **March 21** marks the offset of the this equinox.

2. The Autumn equinox

- It appears when the sun crosses equator giving approximately equal length between day and night.
- It appears to happen when the visible sun moves **south** across the celestial equator on **23rd of September**.
- It marks the beginning of Autumn season.

Solstice

- It is an event when the overhead sun appears to **cross northern or southern points** relative to the celestial equator resulting in **unequal length of days and nights** in the hemispheres.

The Summer Solstice

- On June **21st**, the **northern hemisphere** has maximum tilt towards the sun experiencing **longest daylight of the year**.
- It is the astronomical **first day of summer** in the Northern Hemisphere.
- The sun is at its highest position in the noonday sky, directly above **$23\frac{1}{2}^0$ in the Tropic of Cancer**.

The Winter Solstice

- On **22nd of December**, the day when the **maximum southward** inclination is attained in the Southern Hemisphere..

- In this event the sun travels shortest length causing **longest night and shortest daylight.**
- In the Northern Hemisphere, it occurs when the sun is directly over the **Tropic of Capricorn**, which is located at 23½° south of the equator.

c. Altitude

- Altitude is the **height of location** above the sea level.
- Under normal conditions there is a general **decrease in temperature with increasing elevation.**
- The average rate at which temperature changes per unit of altitudinal change is known as **lapse rate**.
- The lapse rate is limited to the lower layer of the atmosphere (troposphere)

- The normal lapse rate is **6.5°C** per kilometer rise in altitude.
- There are **3 types of lapse rates**:

i. Dry adiabatic laps rate

- An adiabatic lapse rate is the rate at which the temperature of an air parcel changes in response to the **expansion or compression** process associated with a change in altitude.
- When air rises, **it expands** because there is less weight of air upon it.
- Thus, if a mass of dry air at sea level rises to an altitude of about 5486.22meters, the **pressure upon it is reduced by nearly half** and consequently its **volume is doubled**.

- If the upward movement of air does not produce condensation, then the energy expanded by **expansion will cause the temperature of the mass to fall at the constant dry adiabatic lapse rate**.
- The rate of heating or cooling is about **10°C** for every 1000m of change in elevation.
- This rate applies only to **unsaturated air**, and thus it is called the **dry adiabatic laps rate**.
- The rate at which **rising or sinking saturated air changes** its temperature is less than the dry adiabatic rate.
- Prolonged cooling of air invariably produces **condensation**, thereby liberating latent heat.

- Therefore, rising and saturated or precipitating air cools at a slower rate than air that is unsaturated.
- This process is called **wet adiabatic temperature change**.
- The rate of cooling of wet air is approximately **5^0c** per 1000meters ascend.

iii. Environmental lapse rate/Atmospheric lapse late

- This refers to the **actual, observed change** of temperature with altitude.
- The fact that air temperature is normally highest at low elevations next to the earth and decreases with altitude.

- This decrease in temperature upward from the earth's surface normally prevails throughout the lower atmosphere called **troposphere** because of temperature **inversions.**
- The rate of change is ***6.5***^{0}C/1000meters.

5.3.1 Spatio-temporal Distribution of Temperature

- The spatial distribution of temperature in Ethiopia is primarily determined by **altitude and latitude**.
- The location of Ethiopia at close proximity to equator, a zone of maximum insolation, resulted for every part of the country to experience **overhead sun twice a year**.
- However, in Ethiopia, as it is a **highland country**, tropical temperature conditions **have no full spatial coverage**. They are limited to the lowlands in the peripheries.

- Away from the peripheries the land begins to rise gradually and considerably, culminating in peaks in various parts of the country.
- Thus temperature, as it is affected by altitude, **decreases towards the interior highlands.**
- Mean annual temperature varies from over **30^0C** in the tropical lowlands to less than **10^0C** at very high altitudes.
- The **Bale Mountains** are among highlands where **lowest mean annual temperatures are recorded.**
- The **highest mean maximum temperature** in the country is recorded in the **Afar Depression**. Moreover, lowlands of **NW, W** and **SE** Ethiopian experiences mean maximum temperatures of > **30^0C**

- Environmental influences have their own **traditional expressions in Ethiopia** and there are local terms denoting temperature zones as shown in the table below.

Altitude (meter)	***Mean annual Temp (^0C)***	***Description***	***Local Equivalent***
3,300 and above	*10 or less*	Cool	*Wurch*
2,300 - 3,300	*10 – 15*	Cool Temperate	*Dega*
1,500 - 2,300	*15 – 20*	Temperate	*Woina Dega*
500 - 1,500	*20 – 25*	Warm Temperate	*Kola*
below 500	*25 and above*	Hot	*Bereha*

- The **temporal** distribution of Ethiopian temperature is characterized by extremes.
- The major controls determining its distributions are **latitude and cloud cover.**
- However, some parts of the country enjoy a **temperate climate**.
- In the tropics, the daily range of temperature is **higher** and the annual range is **small**, whereas the **reverse is true in the temperate latitudes.**
- In Ethiopia, as in all places in the tropics, the air is **frost free** and **changes in solar angles are small** making intense solar radiation.
- Ethiopia's **daily temperatures are more extreme than its annual averages.**

- Daily maximum temperature varies from a high of more than **37ºC** over the lowlands in northeast and southeast to a low of about **10-15ºC** over the northwestern and southwestern highlands.
- The variation in the amount of solar radiation received daily is small throughout the year.
- In Ethiopia and elsewhere in the Horn of Africa, **temperature shows seasonal variations.**
- For example, months from **March to June** in Ethiopia have records of **highest temperatures**.
- Conversely, **low temperatures** are recorded from **November to February.**
- It is not easy to observe **distinct variation** in temperature between seasons as the sun is always high in the tropics.
- However, there is **a slight** temperature increase in summer.

- **Southern** part of Ethiopia receives highest records of temperature in **autumn and spring** following the relative shift of the sun; whereas in the **northern** part of the country, **summer** season is characterized by higher temperature.
- It has to be noted that **certain seasons should have special considerations.**
- For instance, unlike other parts of Ethiopia, **the southern and southwestern highlands** experience **reduced temperature**.
- This is because the temperature and the amount of energy reaching the surface is directly related with the **directness of the sun.**

5.3.2 Spatio-temporal Distribution of Rainfall

- Rainfall in Ethiopia is influenced by the position of **Intertropical Convergence Zone (ITCZ).**
- The convergence of **Northeast Trade winds and the Equatorial westerlies** forms the ITCZ, which is a **low-pressure zone**.
- The inter-annual oscillation of the surface position of the ITCZ causes a variation in the **Wind flow patterns** over Ethiopia and the Horn.
- Following the **position of the overhead sun**, the **ITCZ shifts north and south of the equator.**
- As the shift takes place, **equatorial westerlies** from the **south and southwest** invade most parts of Ethiopia bringing moist winds.
- The ITCZ shifts towards south of the equator (Tropic of Capricorn) in **January.**

- During this period, the **Northeast Trade Winds** carrying **non-moisture**-laden dominates the region.
- **Afar and parts of Eritrean coastal areas** experience rainfall in this period.
- Following the directness of the sun in **March and September** around the equator, the **ITCZ shifts towards equator**.
- During this time, the **central highlands, southeastern highlands and lowlands** receives rainfall as the **south easterlies** bring moist winds.

Seasonal or Temporal Variability's of rainfall in Ethiopia

➢ The seasonal and annual rainfall variations observed in Ethiopia are the results of the **macro-scale pressure systems and monsoon flows** which are related to the **changes in the pressure system**.

1. Summer season (June, July, August)

- From mid-June to mid-September, majority of Ethiopian regions, except lowlands in Afar and Southeast, receive rainfall during the summer season as the sun overheads north of the equator.

- **High pressure cells** develop on the **Atlantic and Indian Oceans** around the **tropic of Capricorn** although the Atlantic contributes a lot; the Indian Ocean is also source of rainfall.
- During this season, Ethiopia and the Horn of Africa come under the influence of the **Equatorial westerlies (Guinea monsoon) and Easterlies**.
- Hence, these winds are responsible for the rain in this season.

2. Autumn Season (September, October and November)

- In autumn the ITCZ shifts towards the **equator** weakening the equatorial westerlies.
- During this season, the south easterlies from **Indian Ocean showers the lowlands in southeastern part of Ethiopia.**

3. Winter season (December, January and February)

- The overhead sun is far **south of equator**.
- **Northeasterly winds** originating from the **landmass of Asia** dominantly prevail Ethiopian landmass.
- However, it has **no significant coverage** compared to other seasons.
- The northeasterly winds crossing the **Red Sea** carry very little moisture and supplies rain only to the **Afar lowlands and the Red Sea coastal areas.**

4. Spring Season (March, April and May)

- The noonday **sun is shining directly on the equator** while shifting north from south.
- The shift of the ITCZ, results in **longer days and more direct solar radiation** providing **warmer weather** for the northern world.
- In this season, the effect of the **northeast trade wind** is very much **reduced.**
- **Conversely**, the **Southeasterlies** from the **Indian Ocean provide** rain to the **highlands of Somalia, and to the central and southeastern lowlands and highlands of Ethiopia**.

Rainfall Regions of Ethiopia

- Based on **rainfall distribution**, both in **space and time**, four rainfall regions can be identified in Ethiopia and the Horn. These are:

i. Summer rainfall region

- This region comprises **almost all parts of the country**, **except the southeastern and northeastern lowlands.**
- The region experiences most of its rain during summer (*kiremt*), while some places also receive spring (*Belg*) rain.
- The region is divided in to **dry and wet** summer rainfall regions.
- Hence, the **wet** corresponds to the area having rainfall of **1,000mm or more.**
- The **High altitudes** and the **windward side** experience such rainfall amount.

ii. All year-round rainfall region

- It has **many rainy days** than any part of the country.
- It is a rainfall region in the **southwestern part** of the country.
- The wetness of this region is particularly due to the prepotency of **moist air currents of equatorial westerlies called the Guinea Monsoons.**
- Both duration and amount of rainfall **decreases as we move from southwest to north and eastwards.**
- The average rainfall in the region varies from **1,400 to over 2,200mm/year.**

iii. Autumn and Spring rainfall regions

- The region comprises areas receiving rain following the influence of **southeasterly winds**.
- **South eastern lowlands** of Ethiopia receive rain during autumn and spring seasons when both the **north easterlies and equatorial westerlies are weak.**
- About **60% of the rain is in autumn** and **40% in spring**.
- The average rainfall varies from less than 500 to 1,000mm.

iv. Winter rainfall region

- This rainfall region receives rain from the **northeasterly winds**.
- During the winter season, the **Red sea escarpments** and some **parts of the Afar region** receive their main rain.

5.4 Agro-ecological Zones of Ethiopia

- Due to diversified altitude and climatic conditions, Ethiopia possesses five agro-climatic zones (*Bereha, Kolla, Woina Dega, Dega and Wurch*) traditionally been defined in terms of **temperature**.

Zones	Altitude (m)	Mean annual rainfall (mm)	Length of growing periods (days)	Mean annual temperature (^{0}C)	Area share (%)
Wurch (cold to moist)	>3,200	900-2,200	211–365	Below 10	0.98
Dega (cool to humid)	2,300 - 3,200	900-1,200	121–210	≥11.5–17.5	9.94
Weyna Dega (cool sub humid)	1,500 - 2300	800-1,200	91–120	>17.5 – 20.0	26.75
Kola (Warm semiarid)	500 - 1,500	200-800	46–90	>20.0 – 27.5	52.94
Berha (Hot arid)	<500	Below 200	0–45	>27.5	9.39

The Wurch Zone

- The Wurch-zone is an area having altitude **higher than 3,200meters** above sea level and mean annual temperature of **less than 10^{o}C.**
- Mountains having typically fitting characteristics of this zone include mountain systems of **Ras Dashen, Guna, Megezez in North Shoa, Batu, Choke, Abune Yoseph** etc.

Dega Zone

- This is a zone of highlands having relatively higher temperature and lower altitude compared to the *wurch* Zones.
- In Ethiopia, the *Dega*-zone is **long inhabited and has dense human settlement** due to reliable rainfall for agriculture and absence of vector-borne diseases such as malaria.
-

Weyna Dega Zone

- It lies between 1500-2,300meters above sea level
- This zone has **warmer temperature and moderate rainfall** highly suitable for majority of crops grown in Ethiopia. Hence, the zone includes most of the agricultural land.
- It is the **second largest zone covering more than 26%** of the landmass of Ethiopia.
- The *Weyna Dega* zone has also **two growing seasons**.

Kolla Zone

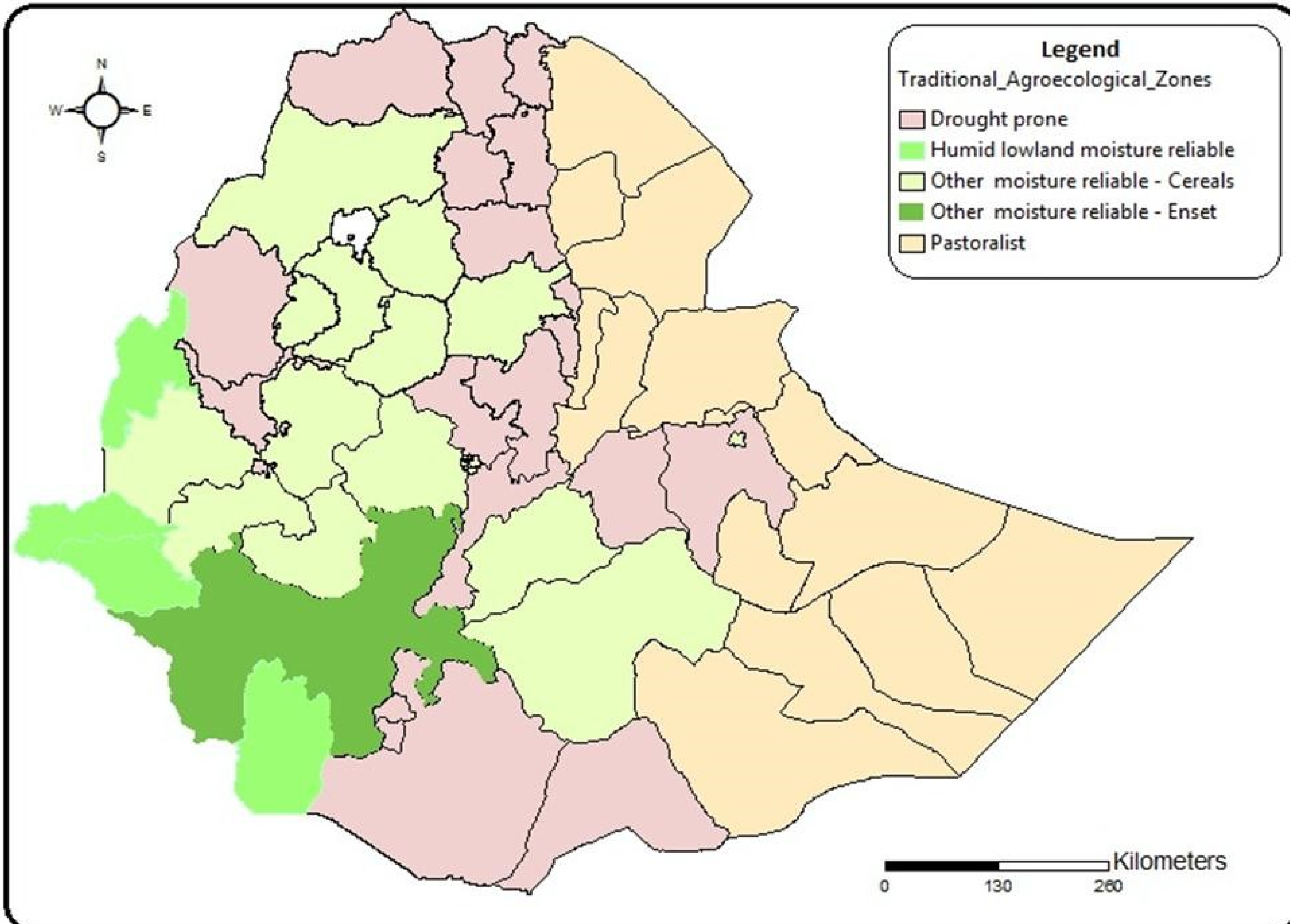
- Peripheries in **south, southeast, west and northeastern** part are mainly in this category making the **first largest zone covering more than 52%**
- It is the climate of the **hot lowlands** with an altitudinal range of 500 to 1500meters above sea level.
- Average annual temperature ranges between **20^{o}C and 30^{o}C**.

- Although mean annual **rainfall is erratic**, it can be as high as **1500mm in the wet western lowlands of Gambella**.

Bereha Zone

- It is the **hot arid climate of the desert lowlands** largely confined to lowland areas with altitude of lower than 500meters.
- Around Danakil depression, the elevation goes below the sea level.
- Its average annual rainfall is less than 200mm, and average annual temperature is over 27.5^{o}C.
- Djibouti, majority of Somalia, and coastal areas of Eritrea are categorized under ***Kolla*** and ***Bereha*** zones.

Fig


N
W
E
S
Legend
Traditional_Agroecological_Zones
Drought prone
Humid lowland moisture reliable
Other moisture reliable - Cereals
Other moisture reliable - Enset
Pastoralist
Kilometers
0
130
260

5.5. Climate Change: Causes, Consequences and Response Mechanisms

- Climate change refers to a change in the state of the climate that can be identified by changes in the **mean and/or the variability of its properties** and that **persists for an extended period**, typically decades or longer.

5.5.1. Current Trends of Climate in Ethiopia

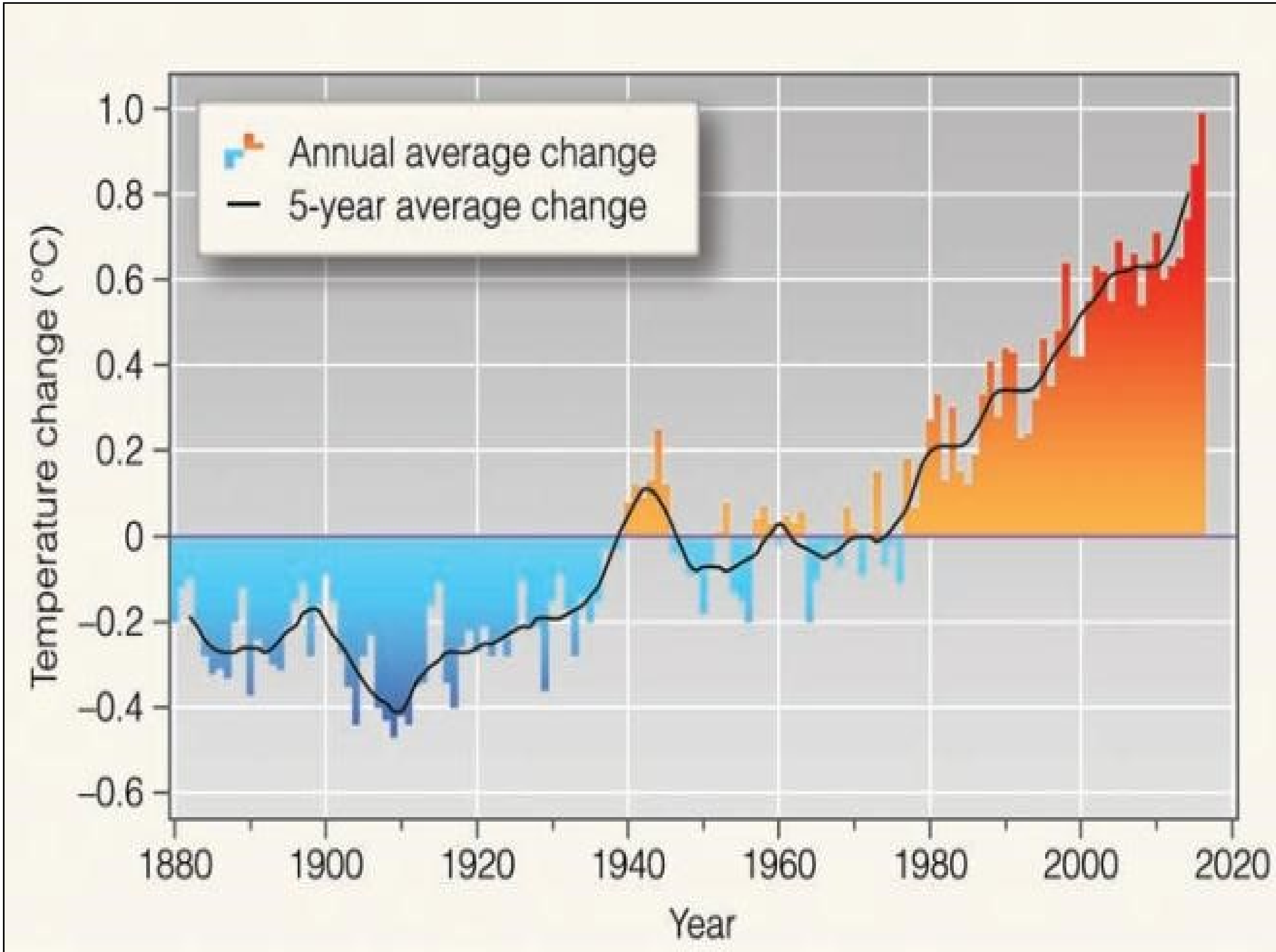
- Besides spatial & temporal variations in different parts of the country, Ethiopian climate experiences **extremes such as drought, flood etc.**
- Ethiopia ranked **5th** out of **184 countries** in terms of its risk of drought.

- In Ethiopia, **12 extreme drought events** were recorded b/n **1900 - 2010**.
- Among the 12, **seven of the drought events occurred since 1980**. The majority of these resulted in **famines**.
- The severe drought of 2015-2016 was exacerbated by the **strongest El Nino** that caused successive **harvest failures and widespread livestock deaths** in some regions.

Trends in Temperature Variability

- In Ethiopia the mean annual temperature has shown **0.2°C to 0.28°C rise per decade** over the last 40-50 years.

- A rise in average temperature of about **1.3°C** has been observed between 1960 and 2006.
- Higher rise in temperature **was noted in drier areas** in northeast and southeast part of the country.
- Notably the **variability is higher in July-September**.
- Consequently, the country's minimum temperature has increased with **0.37°C to 0.4°C per decade**.


Annual average change
5-year average change
Temperature change (°C)
1.0
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
1880
1900
1920
1940
1960
1980
2000
2020
Year

Trends in Rainfall Variability

- Precipitation has remained **fairly stable** over the last 50 years when averaged over the country.
- Rainfall **variability is increasing** (and **predictability is decreasing**) in many parts of the country.
- In some regions, total **average rainfall is showing decline**.
- For instance, parts of **southern, southwestern and south-eastern** regions receiving Spring and Summer rainfall have shown decline by 15-20% between 1975 and 2010.

5.5.2. Causes of Climate Change

A. Natural Causes

➢ **Earth orbital changes**

- Changes in the tilt of the earth at an angle of 23.5° can lead to small but climatically important changes in the strength of the seasons.

▪ **More tilt means** warmer summers and colder winters.

➢ **Energy Budget**

▪ Although the **Sun's energy** output appears constant, **small changes** over an **extended period of time** can lead to climate changes.

• Since the Sun was born, 4.5 billion years ago, the star has been very **gradually increasing its amount of radiation** so that it is now **20% to 30% more intense than it was once.**

➢ **Volcanic Eruptions**

- It releases large volumes of **sulphur dioxide, carbon dioxide, water vapor, dust, and ash** into the atmosphere.
- The release of large volume of gases and ash can increase **planetary reflectivity causing atmospheric cooling**.

B. Anthropogenic Causes

- The warming of earth planet in the past 50 years is **majorly driven** by human activities.

- The **industrial activities** that our modern civilization depends upon have raised atmospheric **carbon dioxide levels** from 280 parts per million to 400 parts per million in the last 150 years.

- **Human induced greenhouse gases** such as **carbon dioxide, methane and nitrous oxide** have caused much of the observed increase in Earth's temperatures over the past 50 years.

- The major gases that contribute to the greenhouse effect include **Water vapor, Carbon dioxide (CO_2), Methane, Nitrous oxide, Chlorofluorocarbons (CFCs).**

- Although **Methane** is less abundant in atmosphere, it is by far more **active greenhouse gas** than carbon dioxide.

5.5.3. Consequences of Climate Change

- In many parts of the world, climate change has already caused **loss of life, damaging property and affecting livelihoods**.
- The impact of climate change is **higher in low income countries**, since they have **limited capacity to cope with the changes**.
- Some of the consequences of the climate change include:

✓Impacts on human health

✓Impact on water resources,

✓Impact on Agriculture, and impact on Ecosystem.

5.5.4. Climate Change Response Mechanisms

- There are **three major response** mechanisms to climate change namely **mitigation, adaptation and resilience**.

Climate Change Mitigation Strategies

- Mitigation measures are those actions that **are taken to reduce and control greenhouse gas emissions changing the climate**.
- Reducing the flow of heat trapping greenhouse gases into the atmosphere, **either by reducing sources of these gases or enhancing the “sinks”** that accumulate and store these gases (such as the oceans, forests and soil).

- There are some climate change mitigation measures that can be taken to avoid the increase of **pollutant emissions**.

❖Practice Energy efficiency,

❖Increase the use of renewable energy such as solar energy

❖Efficient means of transport implementation: electric public transport, bicycle, shared cars etc.

Climate change adaptation strategies

- Adaptation is simply defined as **adapting to life in a changing climate**.
- It involves **adjusting to actual or expected future climate**.

- The goal is to **reduce our vulnerability** to the harmful effects of climate change such as extreme weather events or food insecurity.
- It also encompasses making the most of any **potential beneficial opportunities** associated with climate change

❑Some of the major **climate change adaptation** strategies are:

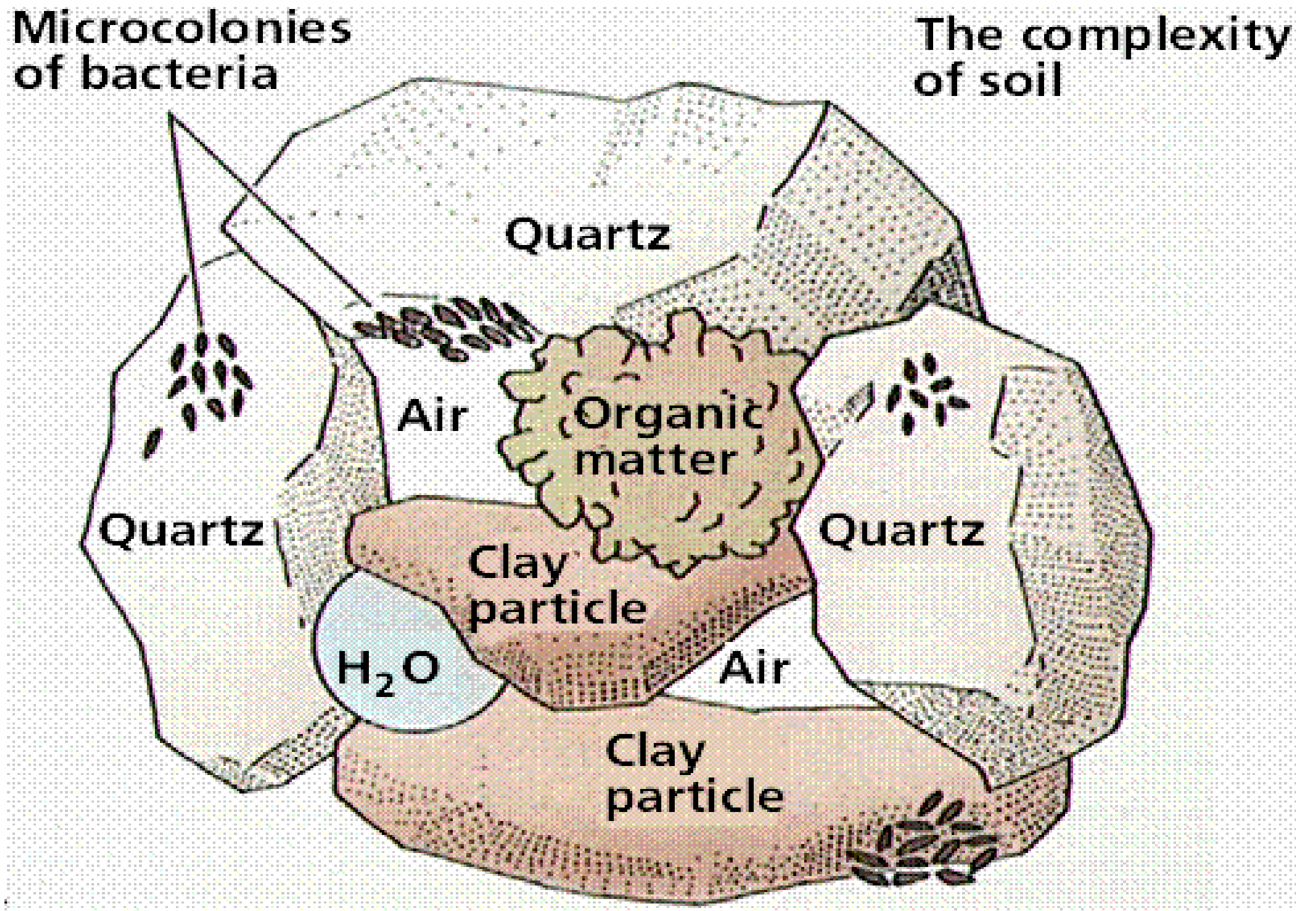
- Building flood defenses,
- Plan for heat waves and higher temperatures,
- Installing water-permeable pavements to better deal with floods and storm water,
- Improve water storage,

- Landscape restoration and reforestation,
- Flexible and diverse cultivation to be prepared for natural catastrophes,
- Preventive and precautionary measures such as evacuation plans, health issues, etc.

CHAPTER SIX

SOILS, NATURAL VEGETATION AND WILDLIFE RESOURCES OF ETHIOPIA AND THE HORN

- **Soil** is composition of **mineral particles (45%), organic matter (5%) air (25%) and water (25%)** that together support life.



•This variability reflects primarily **the parent material** from which the soil was formed over very long periods of time and the **environment** in which the soil has developed.

How soil is formed?

•**Soil formation is a long-term process:** It could take **several thousands of years** to form a single stratum of soil.

•As it is a complex **mixture of several constituents**, **its formation is also more complex.**

•The formation of a particular type of soil depends on **parent material, climate, topography, living organism and length of time.**

•**Weathering** disintegrates the inorganic substances (rocks) of soils.

•It is the breakdown of rocks at the Earth's surface, by **the action of rainwater, extremes of temperature, and biological activity** (Decomposing animals and plants).

- There are 3 types of weathering involving in soil formation.

A. **Mechanical (Physical) weathering**

B. **Biological weathering**

C. **Chemical weathering**

A. Mechanical (Physical) Weathering

•Physical disintegration causes **decrease in size without appreciably altering composition.**

•Differential stresses due to **heating & cooling**/ expansion of ice break the rock.

•**Abrasion** (erosion by friction) due to water containing sediment or wind carrying debris is another type of physical weathering.

B. Biological Weathering

•The process of biological weathering involves the weakening and subsequent disintegration of rock by **plants, animals and microbes**.

•**Roots of plant** can exert pressure on rock.

•Although the **process is physical**, the **pressure is exerted by a biological process** (i.e., growing roots).

•**Microbial activity** breaks down rock minerals by **altering the rock's chemical composition**, thus making it more **susceptible to weathering**

C. Chemical Weathering

•**Chemical weathering** involves the **modification of the chemical** and mineralogical composition of the weathered material.

•A number of different processes can result in chemical weathering.

•The most common chemical weathering processes are hydrolysis, oxidation, reduction, hydration, carbonation, and solution.

•In most cases, **the minerals in the parent materials** are also found in the soils, which are formed from the disintegration and decomposition of the rock.

•However, this is **not true of alluvial soils**, which are transported from one place to the other by agents like running water.

Properties of Soil

•**Soils have two basic properties: Physical and chemical**

A. Physical properties

•**Soil physical properties** are influenced by **composition and proportion** of major soil components.

•Properties such as **texture** (Different sized mineral particles give soil its texture: Sand, Silt and Clay), **structure**, **porosity** etc. are categorized under physical soil properties.

•These properties affect **air and water movement in the soil**, and thus the **soil's ability** to function.

B. Chemical Properties

•**Soil chemistry** is the interaction of various **chemical constituents** that takes place among soil particles and in the water retained by soil.

•**Soil properties** like **availability of minerals, electrical conductivity, soil pH**(pH=7, Neutral, pH<7, Acidic and pH>7, Alkaline) etc ,,,,,,

•**Soil chemical properties** affect **soil biological activity** and indirectly the **nutrient dynamics**.

6.1.1. Major Soil Types in Ethiopia

- Soils of Ethiopia are basically derived from **crystalline, volcanic and Mesozoic sedimentary rocks.**
- FAO has identified **18 soil** associations in Ethiopia
- Out of the major soils, 11 soil associations cover about 87.4 percent of the land area.
- The **six major groups** of soils in Ethiopia are discussed under the following points:

A. Environmental condition i.e. parent material, climatic conditions, topography, the way they were formed.

B. Characteristic i.e. significant chemical and physical properties

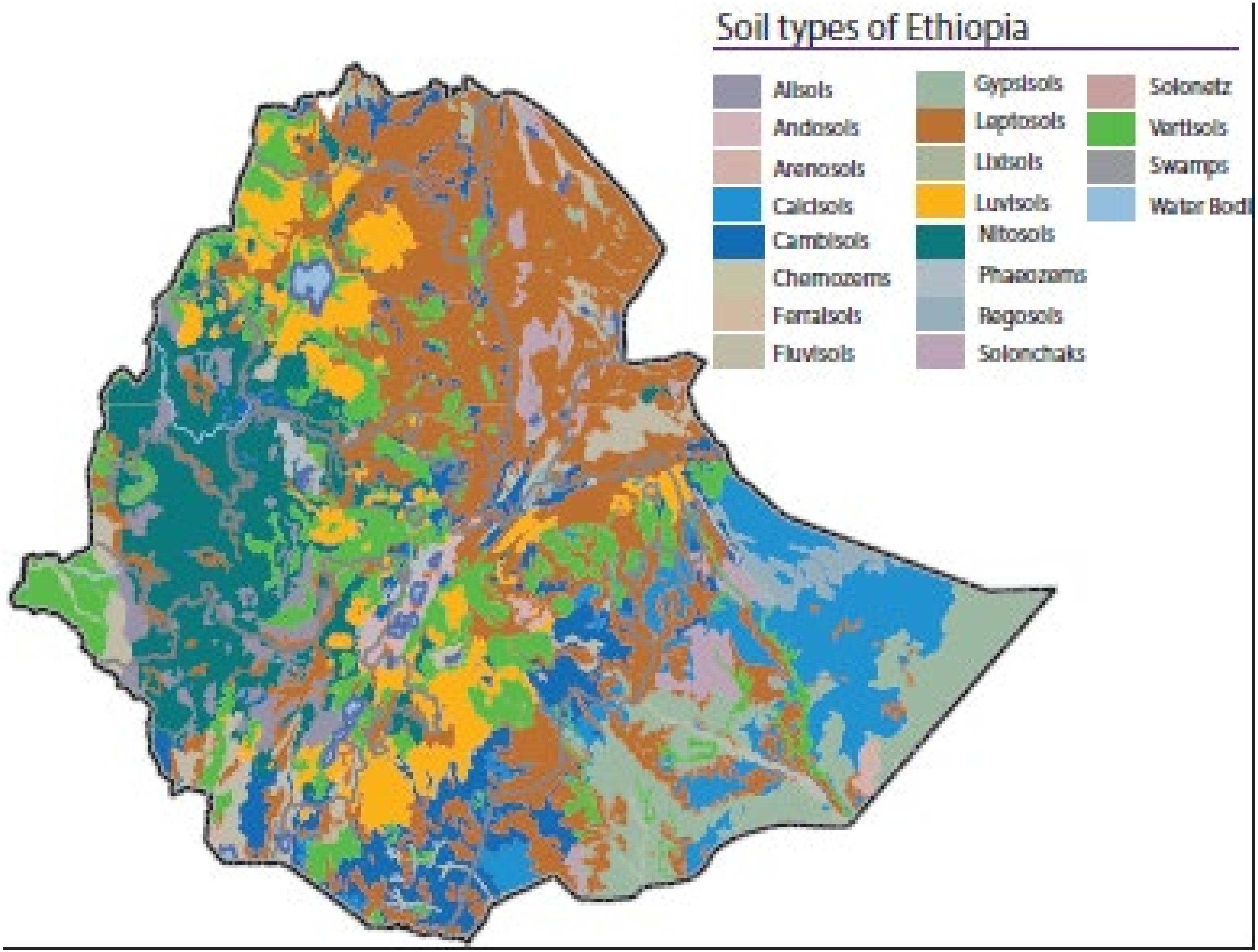
C. Agricultural suitability in relation to texture, structure, topography, moisture-storage capacity, etc.

D. Occurrence: general location of the soil types.

The six major groups of soils in Ethiopia

1. Nitosols and Acrisols
2. Vertisols
3. Lithosols, Cambisols and Regosols
4. Xerosols, Yermosols and Solanchaks
5. Fluvisols and
6. Luvisols

Major Soil Types in Ethiopia……



1. Nitosols and Acrisols

Nitosols

- Nitosols develop on **gently sloping** ground.
- Their parent materials include **trap series volcanic, volcanic ash, and even metamorphic rocks.**
- They are **strongly weathered soils** but far more productive than most other tropical soils.
- They are basically associated **with highlands with high rainfall** and they were, probably, formed on **forest covered areas** originally.

- Due to the high rainfall, there is considerable **soil leaching** which makes the nitosols to be **poor in soluble minerals** like **potassium, calcium** etc.; and **rich in non-soluble minerals** like **iron and aluminum**.
- The **reddish-brown color** of these soils is because of **high concentration of iron** (ferric) oxides due to leaching.
- They are now widely found on **cultivated areas and on mountain grasslands.**
- Nitosols are dominantly **found in western highlands** (Wellega), southwestern highlands (Kaffa, Illuababora), **Southern highlands**, **Central highlands, and Eastern highlands.**

Nitosols

Acrisols

- Acrisols are one of the most **inherently infertile soils** of the tropics, becoming **degraded chemically and organically very quickly when utilized.**
- Acrisols have very **low resilience to degradation** and moderate sensitivity to yield decline.
- In Ethiopia, it has **lost most of the base nutrients** and are characterized by **low productive capacity**.
- Acrisols are found along with nitosols mostly in **some pockets of southwestern highlands** of Ethiopia where there is high rainfall.

Acrisols

2. Vertisols

- Vertisols are **heavy clay soils** (expansive clay minerals)with a high proportion of **swelling** clays when wet, and **cracks** when dry.
- These soils are **extremely difficult to manage** (hence easily degraded), but has **very high natural chemical fertility**.
- Vertisols mostly develop on **volcanic plateau basalt, trachyte and pyroclastic materials, sedimentary rocks, colluvial slopes and alluvial plains.**
- The vertisols are soils of **highlands and moderate climates**.

- In Ethiopia, they are commonly found in parts of **Northwestern, Central and Southeastern highlands** (especially in Gojjam, Shewa, Arsi, Bale and central Hararghe).

3. Lithosols, Cambisols and Regosols

- These soils are mostly found in **rugged topography and steep slopes**.
- There is **little evidence of pedogenic processes** (soil forming processes).
- As a result, they are **young, shallow and coarse textured** and so have **low water holding capacity**.
- In addition, they are **found in areas of low rainfall.**
- So, most of the areas covered by these soils have **limited agricultural use.**
- They are, in most cases, **left under the natural plant cover and used for grazing.**

Lithosols, Cambisols and Regosols…..

- By large, these soils are found in different parts of **rugged and steep slopes** of **Central Highlands**, on the **Rift Valley Escarpments** and highlands of **Western Hararghe**.

- Regosols and Lithosols are also found in the **Danakil and eastern Ogaden.**

4. Xerosols, Yermosols and Solanchaks

- These are soils of **desert or dry steppe soils** majorly available in **arid and semiarid areas.**
- Though the degree may vary, desert soils are characterized by **high salt** content and **low organic content**, because of the **scanty vegetation.**
- Generally speaking, these soils **have poor humus content and nitrogen**, but are **rich in phosphorus and potash** and **can be very fertile if irrigated.**
- **Xerosols are soils of the deserts** and are extremely **subjected to wind erosion and concentration of soluble salts.**
- Yermosols are **even drier** and **more problematic** than Xerosols.

Xerosols, Yermosols and Solanchaks…..

- Solanchaks are **saline soils** which develop in areas of **high evaporation and capillary action**.
- **Badly managed irrigation schemes** may turn soils into solanchaks.
- In Ethiopia, Xerosols are found in **Ogaden and northeastern escarpments**, whereas the Yermosols and Solanchaks cover the **Ogaden and Afar plains**.
- The Solanchaks are majorly located in **salty plains of Afar**.

5. Fluvisols

- Fluvisols develop on **flat or nearly flat ground**, on **recent alluvial deposits.**
- These soils are associated with **fluvial (river), marine (sea) and lacustine (lake) deposits.**
- These are soils formed due to **deposition of eroded materials** from highlands.
- The deposition takes place in **depressions**, lower valleys and lowlands.
- **Lower regions of rivers** like Omo, Awash, Abay and the plains of Akobo and Baro Rivers are home for fluvivsols.
- **Lakes region** (main Ethiopian rift) is also characterized by fluvisols.

•Fluvisols are **highly variable, but much prized for intensive agriculture because:** they develop on flat ground, deposition sites,

✓They are associated with **rivers and ground water**, making them important for **large-scale irrigation** and

✓They are **fertile and their fertility is always renewed as a result of deposition of new soil materials**

6. Luvisols

- Luvisols develop mainly in areas where **pronounced wet and dry seasons occur in alternation**.
- Where **leaching is not very high**, they are found in **association with nitosols.**
- Luvisols have **good chemical nutrients** and they are **among the best agricultural soils** in the tropics. So, they are intensively cultivated.
- However, when luvisols are found on **steep slopes** (stony) and **on flat areas (waterlogged)** they are **avoided and left for grazing**.

- In Ethiopia, places with luvisols include **Lake Tana area, parts of Northern, Central and Eastern Highlands and Southern lowlands**.

6.1.2. Soil Degradation: is defined as a **change in any or all of soil status** resulting in a **diminished capacity of the ecosystem** to provide goods and services.

- It could also be the **deterioration of the physical, chemical and biological properties of soil.**
- It is a **critical and growing global problem**. It is a major concern for at least **two reasons**.

A. soil degradation **undermines the productive capacity of an ecosystem**

B. It **affects global climate** through alterations in **water and energy balances and disruptions in cycles of carbon, nitrogen, sulfur**, and other elements

There are three major types of soil degradation. These are:

i. Physical Degradation: refers to the **deterioration of the physical properties of soil**. This includes:

A. **Compaction:** densification of soil is caused by the elimination or reduction of **structural pores**.

- Soils prone to compaction are **susceptible to accelerated runoff and erosion.**

B. Soil erosion: is a **three-phase process** consisting of the **detachment of individual soil particles, transportation and deposition**.

- The **continuous strike of soil surface by rain droplets** considerably weakness the soil and makes susceptible to erosion.

- When **sufficient amount of water accumulates**, the soil will begin to **move towards lower slope** until the erosive agent loses its energy.
- Erosion of **topsoil by wind and water exceeds soil formation at an alarming rate.**
- Obviously for countries like Ethiopia where **agriculture plays the dominant role in the economy and livelihood of the people**, the causes, consequences and possible ways of minimizing soil erosion require serious consideration.
- In Ethiopia, an **estimated average of 42 tons per hectare of soils is eroded annually.**

ii. Biological Degradation

- **Reduction in soil organic matter content, decline in biomass carbon, and decrease in activity and diversity of soil fauna** are ramifications of biological degradation.
- Because of the **prevailing high soil and air temperatures**, biological degradation of soil is more **severe in the tropics** than in the temperate zone.
- It can also be caused by **indiscriminate and excessive use of chemicals and soil pollutants.**

iii. Chemical Degradation

- **Nutrient depletion** is a major cause of chemical degradation.
- In addition, **excessive leaching of cat-ions** in soils with low-activity clays causes a **decline in soil pH** and **a reduction in base saturation**.
- Chemical degradation is also caused by the buildup of some **toxic chemicals** and an elemental imbalance that is injurious to plant growth.

Causes of Soil Degradation

- Soil degradation may result from **natural & human-induced causes**.
- Topographic and climatic factors such as steep slopes, frequent floods and tornadoes, storms and high-velocity wind, high-intensity rains and drought in dry regions are among the **natural causes**.
- Deforestation and overexploitation of vegetation, overgrazing, indiscriminate use of agrochemicals & lack of soil conservation practices, over extraction of ground water are some **anthropogenic causes of soil degradation.**

6.1.3. Soil Erosion Control Measures

- The aim of soil conservation is **to reduce erosion** to a level at which the maximum sustainable level of agricultural production, grazing or recreational activity can be obtained from an area of land without unacceptable environmental damage.
- Since **erosion is a natural process**, it **cannot be prevented**. But it **can be reduced** to a maximum acceptable level or soil loss tolerance.
- **We have two major soil erosion control mechanisms:**

A. Biological

B. Physical

A. **Biological Control measures**

- These types of soil erosion control mechanisms include **vegetative strips, plantation, and reforestation.**
- Biological controls can **prevent splash erosion**, **reduces the velocity of surface runoff, increases surface roughness** which **reduces runoff** and **increases infiltration**, and etc.

Figure 1. Farmland protected Vetiver with terrace in Metu, Ethiopia

B. Physical control measures

- Are used to **control the movement of water & wind** over the soil surface.

•The major types of physical erosion control measures commonly applied in Ethiopia includes **terracing, check dams, gabion, trenches, contour ploughing, soil bunds etc**.


a)
a)

6.2. Natural Vegetation of Ethiopia

6.2.1. Major Natural Vegetation Types of Ethiopia

- Taking altitude into consideration it is possible to broadly classify the vegetation belts of Ethiopia into the following five groups.

1.Afro-alpine and sub-afro alpine Region

2.Forest Region

3.Woodland Savannah Region

4.Steppe Region

5.Semi-desert Region

1.Afro-alpine and sub-afro alpine Region

- Ethiopia has the largest extent of Afro-alpine and sub afro-alpine habitats in Africa.
- This vegetation type, also known as high mountain vegetation is similar to the Alpine vegetation in temperate regions.
- These ecosystems are found on mountains having an elevation ranging between 3,200 and 4,620 meters above sea level.
- The Afro-alpine habitat covers nearly 1.3% of the total landmass of Ethiopia.
- **The Afro-alpine** region is found at very high altitudes (4,000 – 4,620 m).

Afro-alpine and sub-afro alpine Region…..

- Like any other landform in Ethiopian, the climate of Afro-alpine ecosystems is controlled by latitude and altitude.

- The annul precipitation which ranges between 800 and 1,500 mm, is mostly in the form of sleet or snow.

- Temperature records of $0^{o}C$ and below are widely experienced in these ecosystems.

- Soils in this ecosystem are mostly shallow and eroded. The Bale and Semein mountains are typical examples of afro-alpine vegetations.

- Compared to the Afro-alpine, the **Sub-afro-alpine** region is found at a lower elevation, roughly between 3,300 and 4,000 meters.

- As a result, the plants in this region are adapted to somewhat less extreme environment than the Afro-alpine.
- Vegetation in the Afro-alpine region consists of tussock grasslands, scrub, scattered mosses and lichens while the Sub-afro alpine region is dominated by woodland, often degraded to scrub stages and also wet grasslands.
- Lobelia rhynchopetalum (giberra) and Erica arborea (Asta) are some of the dominant species in the Afro-alpine and Sub-afro alpine regions respectively.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2. Forest Region

- Forest is a complex ecosystem consisting predominantly of **trees that shield earth and support numerous life forms**.
- Not all forests are similar in terms of **species composition, structure and physiognomy.**
- In any geographical region, environmental factors such as **climate, soil types, topography and elevation** determine the types of forests.
- In Ethiopia, forests are found at **different elevations**, 450 to 3,500m in humid parts and 2,300 to 3,300 m in most arid parts.
- Moreover, forests are characterized by variation in mean annual rainfall that range between 200 and 2,200mm.

- These wide variations in rainfall and altitude result in **two broad classification of forests: Highlands and Lowland forests**.
- Highland forests include Hagenia Abyssinia (Kosso), Juniper procera (Tid), Arundinaria Alpina(Kerkha), Podocarpus falcatus (Zigba), Aningeria adolfi friedericii (Keraro) and Olea Africana (Weyra) forests; while Baphia are classified as lowland forests.
- ***Baphia*** is a small **genus of legumes** that bear simple leaves
- Moreover, there are also **Gallery (Riverine) Forests**.
- Gallery (Riverine) forests are forests that stretch along the **banks of the lower courses of rivers.**

Forest Region….

- Riverine forests are classified as lowland forests and are found in some places such as the banks of Awash, Wabishebelle, Ghenale etc.
- Dominant species include Ficus sur (sholla) and different kinds of Acacia trees.

3.Woodland Savannah Region

- Like the forests, the woodland savannahs are also found in areas of **wide altitudinal ranges** (250 to 2,300 m).
- Although the mean annual rainfall ranges between 200 and 1,400 mm, the large part of this region is found at a lower elevation and in a drier environment.



- The plants in the woodland savannah are known for their **xeromorphic characteristics** like shading of leaves during the dry season.
- Vegetation types with intermediate characteristics between savannahs and woodlands are **shrub lands and bush lands**.
- Woodland savannah region can be broadly classified into **three** divisions:

✓**Juniper Procera** (Tid) is dominant species for both the Junipers Forests and Junipers Woodlands.

- The difference is in height: 3 - 45 meters tall in the forests and 10 -15 meters in the woodlands.

- ✓**Acacia woodlands** are dominated by both trees and shrubs, which belong to the same genus 'Acacia'.

  E.g. Acacia etbaica(Grar), Acacia Mellifera (Konter).

- ✓**Mixed deciduous woodlands**: As the name implies, most of the trees in mixed deciduous woodlands **shed their leaves** in the dry season.

Woodland savanna type	Altitude(m)	MeanAnnual RF (mm)	Growing season(No. of months)	Dominant species
Junipers woodland	1,350-2,200	500-900	4-8	Junipers procera
Acacia woodland	250-2,300	200-1,000	1-9	Acaci aetabica(Grar)
Mixed deciduous woodland	300-1,300	800-1.400	5-12	Mixed trees

4. Steppe and Semi Desert Regions

- These are regions in the **arid and semiarid** parts of the country where the temperature is very high and the rainfall very low.
- Both are found at **low elevations**, the steppe at elevations of 100 to 1,400 m above sea level and the semi-deserts at 130 meters below sea level to 600 meters above sea level.
- The steppe gets a mean annual rainfall of 100 to 550 mm as compared to 50 to 300 mm for the semi desert areas.
- Growing period lasts up to 2 months for the steppe and a maximum of one month for the semi-deserts.
- Even though there is a variation in the degree of alkalinity and salinity; soils in both regions are generally alkaline and saline.

Steppe and Semi Desert Regions….

- In these regions xerophytic (i.e. drought-resisting plants) are the dominant vegetations.
- Xerophytic plants such as short shrubs, scattered tufts of grass species and a variety of acacias are some of the examples.
- Where there are moist soils, rich vegetation of acacia and palm trees may be observed.
- Trees are normally restricted to fringes along watercourses.



6.2.2. Natural vegetation Degradation

- A century ago, forests covered **about 40 percent** of the total land area.
- For the last few decades, forests have been **cleared for different reasons**.
- Major causes for the gradual disappearance of the natural vegetation in Ethiopia are:

✓Clearing of forests for cultivation

✓Timber exploitation practices

✓Charcoal burning and cutting for fuel

✓Extensions of coffee and tea production areas

✓Overgrazing

✓Expansion of settlements both rural and urban, and Clearing for construction.

6.2.3. Natural Vegetation Conservation

- Conservation of biodiversity is protection and management of biodiversity so as to **maintain at least its current status and derive sustainable benefits** for the **present and future generation**.

- **There are three main approaches of biodiversity conservation:**

✓**Protection:** through designation and management of some form of protected area. Protected areas include sanctuaries, national parks, and community conservation areas.

✓**Sustainable forest management:** involving sustainable harvesting of forest products to provide a source of financial income

- **Restoration or rehabilitation:** is the process of assisting the recovery of a forest ecosystem that has been degraded, damaged, or destroyed
- This may involve the re-establishment of the characteristics of a forest ecosystem, such as composition, structure, and function, which were prevalent before its degradation.

6.3. Wild Life/wild animals in Ethiopia

- Generally speaking, the main wild life concentrations in the country occur in the **southern and western parts**.
- The wild animals in Ethiopia can be classified into five major groups:

1. Common wild animals (those animals that are found in many parts of the country (e.g. hyenas, jackals)

2. Game (lowland) animal, (which include many herbivores like giraffes, wild asses, zebras etc. and carnivores like lions, leopards, and cheetahs)

3. Tree animals or arboreals (which include monkeys, baboons)

4. A variety of birds in the Rift Valley lakes

5. Rare animals (Gelada baboon and Semien fox) scattered in highlands; (walia- ibex in the Semien Massifs, Nyala in the Arsi Bale massifs).

6.3.1. Wildlife Conservation: Wild animals can be used for:

- ✓ scientific and educational researches (valuable information for medical purposes and environmental studies)
- ✓ physical and mental recreation (aesthetic value)
- ✓ promotion of tourism (economic value)
- ✓ its potential for domestication
- ✓ Maintaining ecological balance

- To prevent the destruction of wildlife a total area of nearly 100,000 square kilometers of national parks, sanctuaries, community conservation areas, botanical gardens, wildlife reserves etc. have been established in different part of the country.

Hence in Ethiopia there are:

- **21 major national parks**
- 2 major wildlife sanctuaries,
- 3 wildlife reserves,
- 6 community conservation areas,
- 2 wildlife rescue centres,
- **22 controlled hunting areas,**
- 2 botanical gardens, and
- 3 biosphere reserves

Table 6.2: National Parks of Ethiopia

Name	Region	Year established	Area in sq.km
Kafeta Shiraro	Tigray	1999	5000
Semien Mountains	Amhara	1959	412
Alatish	Amhara		
Bahir Dar Blue Nile River Millennium	Amhara	2008	4729
Borena Saynt	Amhara	2008	4325
Yangudi-Rassa	Afar	1969	4731
Awash	Oromiya & Afar	1958	756
Dati Wolel	Oromiya	2010	1031
Bale Mountains	Oromiya	1962	2200
Yabello	Oromiya	1978	1500

Name	Region	Year established	Area in sq.km
Abijata Shala	Oromiya	1963	887
Arsi Mountains	Oromiya	2012	
Geralle	Somali	1998	3558
Gambella	Gambella	1966	4650
Nechsar	SNNPR	1966	514
Omo	SNNPR	1959	3566
Mago	SNNPR	1974	1947
Maze	SNNPR	1997	202
Gibe Sheleko	SNNPR	2001	248
Loka Abaya	SNNPR	2001	500
Chabra Churchura	SNNPR	1997	1190

Spatial distribution of National Parks of Ethiopia

- Even though the number and the predominant animals may vary, many of the national parks in Ethiopia have different turnovers of animals.
- These include buffaloes, zebras, lions, elephants, ostriches, giraffes, oryx, African wild asses, etc.
- Some of the national parks are unique in their wild animals they have. E.g.

1. Abiyatta-Shalla lakes National Park is predominantly bird sanctuary. Important bird species include the flamingos and pelicans.
2. Omo, Mago, and Gambela National Parks have hippopotamus and crocodiles in rivers and lakes.
3. Semien and Bale Mountains National Parks have rare animals like Walia ibex, Semien fox, gelada baboon and Nyala.

6.3.2. Challenges of wildlife conservation in Ethiopia

- Protected areas were created to protect the major biodiversity.
- However, it is a sad fact that these ecologically fundamental resources are usually undervalued and are under threat from various dimensions.
- Here are some of the **major challenges** that Ethiopian protected areas are facing;

✓Limited awareness on the importance of wild life

✓Expansion of human settlement in protected areas.

✓Conflict over resource

✓Overgrazing (fodder and wood)

✓Illegal wildlife trade

✓Excessive hunting

✓Tourism and recreational pressure

✓Mining and construction material extraction

✓Forest fire

I thank you!!!